

Politecnico di Torino

Politecnico di Torino

Master's Degree

A.Y. 2025/2026

Graduation Session 12 2026

Real-World Guided Depth Super-Resolution via Metric Reconstruction and Monocular Depth Guidance

Supervisors:
Bartolomeo Montrucchio
Antonio Marceddu

Candidate:
Lorenzo Mascia

Abstract

Depth sensors are now integrated in many everyday devices, from smartphones to robots and augmented reality headsets. To keep cost, size and power consumption low, these sensors usually provide depth maps with a much lower resolution than the color images captured by the same device. Guided Depth Super-Resolution (GDSR) addresses this gap by using the high-resolution RGB image to reconstruct a high-resolution depth map from the low-resolution measurement.

Most recent methods for this task are developed in a simplified setting, where the low-resolution depth map is obtained by downsampling the high-resolution ground truth. This provides aligned training pairs, but does not account for the errors introduced by real depth sensors. In practice, depth measurements can contain noise, missing values and distortions that differ from those of the reference sensor.

This thesis studies guided depth super-resolution using real sensor data, focusing on the difficulties of preparing suitable training pairs and reconstructing accurate depth maps with RGB guidance.

As a first step, the Redwood Indoor LiDAR-RGBD dataset is studied as a possible source of real training pairs. A preprocessing pipeline is developed to align measurements from a consumer RGB and Depth (RGB-D) camera with Light Detection and Ranging (LiDAR) geometry and generate high-resolution depth targets. Experiments on this dataset reveal problems related to differences between sensors, geometric alignment, incomplete depth coverage, and calibration.

Building on these findings, this thesis proposes the Boundary-Aware Fusion Network (BAF-Net), a new guided depth super-resolution architecture designed for real-world depth super-resolution. The network first reconstructs a metric depth map from the sensor measurements. It then refines this estimate using RGB information and monocular depth priors obtained from Depth Anything V2. The architecture uses separate feature extraction branches and multi-scale fusion to combine these sources of information. This approach allows the network to recover finer geometric details and better-defined object boundaries using the monocular prior, while preserving

the metric information provided by the sensor.

BAF-Net is evaluated on the Time-of-Flight Depth Super-Resolution (TOFDSR) dataset using a real-world protocol, where the low-resolution depth maps are acquired by a physical sensor instead of being generated through synthetic downsampling. The selected configuration achieves a Root Mean Squared Error (RMSE) of 4.011 cm on the official test set, reaching state-of-the-art performance. This corresponds to a reduction of approximately 43% in RMSE and improves on the previously reported results obtained under the same protocol.

More accurate depth and sharper object borders are useful wherever a device must understand the 3D structure of a scene. In augmented reality, they allow virtual objects to interact correctly with real-world geometry; in robotics, they help locate objects a robot must reach or avoid. Since the method works with the small, low-cost sensors already present in consumer devices, it may also make reliable 3D sensing more affordable for assistive technologies, such as navigation aids for visually impaired people, and for domestic robots.

Acknowledgements

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Tristique senectus et netus et malesuada fames ac turpis. Pretium nibh ipsum consequat nisl vel pretium lectus quam. Urna molestie at elementum eu facilisis sed odio morbi quis. Sed egestas egestas fringilla phasellus faucibus scelerisque eleifend. At in tellus integer feugiat. Mauris rhoncus aenean vel elit scelerisque mauris pellentesque pulvinar. Commodo sed egestas egestas fringilla. Nunc lobortis mattis aliquam faucibus purus in massa. Facilisis magna etiam tempor orci eu lobortis elementum nibh. Elementum curabitur vitae nunc sed velit dignissim. Neque volutpat ac tincidunt vitae. Massa id neque aliquam vestibulum morbi blandit cursus risus at. Porta non pulvinar neque laoreet suspendisse interdum consectetur. Turpis in eu mi bibendum. Ut tristique et egestas quis ipsum suspendisse. Integer quis auctor elit sed vulputate mi sit amet. Viverra nam libero justo laoreet sit amet cursus.

Table of Contents

List of Figures VIII

1 Introduction 1

1.1 Depth Maps and Depth Sensing Technologies 1

1.1.1 Depth Maps 1

1.1.2 Depth Acquisition Technologies 2

1.1.3 RGB-D Cameras and LiDAR 3

1.1.4 Applications of Depth Maps 3

1.2 Depth Super-Resolution 4

1.3 RGB-Guided Depth Super-Resolution 5

1.4 Thesis Motivation and Objectives 6

1.5 Thesis Organization 7

2 Background and Related Work 8

2.1 Deep Learning for GDSR 8

2.1.1 Supervised Learning 8

2.1.2 Convolutional Neural Networks 9

2.1.3 Encoder–Decoder Architectures 9

2.1.4 Residual Learning 10

2.1.5 Attention and Vision Transformers 10

2.1.6 Loss Functions 11

2.2 Approaches to Guided Depth Super-Resolution 11
2.2.1 Filtering-Based Methods 12
2.2.2 Optimization-Based Methods 12
2.2.3 Learning-Based Methods 13
2.2.4 Deep Learning Architectures for GDSR 14
2.3 Synthetic and Real-World DSR Protocols 15
2.4 Real-World Guided Depth Super-Resolution 16
2.5 Depth Foundation Models as Geometric Priors 17
2.6 Common GDSR Benchmarks 18

3 Evaluation of Redwood Indoor LiDAR-RGBD 20
3.1 Dataset and Experimental Formulation 20
3.2 LiDAR-to-Camera Registration 22
3.3 LiDAR Target Generation 24
3.3.1 Visibility and Occlusion Filtering 25
3.3.2 Raw Targets, Densification and Validity Masks 25
3.3.3 Partial LiDAR Coverage 26
3.4 Construction of the $\times 4$ Input 27
3.5 Experimental Protocol 28
3.6 Baseline Networks 29
3.7 Cross-Domain Evaluation 29
3.8 Retraining on Redwood 30
3.8.1 Effect of Denser Temporal Sampling 30
3.8.2 Qualitative Comparison 31
3.9 Resolution Loss and Cross-Sensor Disagreement 32
3.10 Limitations of Redwood as a GDSR Training Benchmark 33

4 Real-World GDSR Experimental Protocol 34
4.1 Real-World GDSR Protocol . . . 34
4.2 TOFDSR Dataset . . . 34
4.3 Input and Ground-Truth Preparation . . . 35
4.4 Evaluation Protocol . . . 35
4.4.1 Validation and Model Selection . . . 36
4.4.2 Protocol Verification . . . 36

5 BAF-Net 37
5.1 Design Overview . . . 37
5.2 Inputs and Normalization . . . 38
5.3 Building Blocks . . . 40
5.4 Network Architecture . . . 41
5.4.1 Depth Branch and Metric Base . . . 42
5.4.2 Guidance Branch . . . 43
5.4.3 Multi-Scale Fusion and Guided Decoder . . . 43
5.5 Properties of the Design . . . 43
5.6 Loss Function . . . 44
5.7 Implementation and Training Details . . . 45
5.8 Summary . . . 46

6 Experimental Results 47
6.1 Experimental Setup . . . 47
6.2 Comparison with the State of the Art . . . 48
6.3 Error Distribution . . . 49
6.4 Ablation Study . . . 50
6.5 Monocular Prior and BAF-Net . . . 52
6.6 Qualitative Results . . . 53

6.6.1 Comparison with DORNet 56
6.7 Summary 59

Bibliography 60

List of Figures

1.1 RGB image (left) and corresponding depth map (right). The depth map provides a pixel-wise representation of scene geometry, where each value is associated with the distance from the sensor to the observed surface. 2

2.1 Synthetic and real-world DSR protocols. (a) In the synthetic protocol, the low-resolution input is generated from the reference depth with a known degradation, so input and target differ only in resolution. (b) In the real-world protocol, input and reference are acquired by two different sensors, so the input also contains the errors of the low-resolution sensor. 16

3.1 Examples from the five Redwood Indoor LiDAR-RGBD scenes used in this work. The top row shows depth maps acquired by the consumer RGB-D sensor, while the bottom row shows the corresponding RGB images. The depth ranges reported below each scene indicate the approximate interval visualized in each depth map. 21

3.2 Overview of the pipeline used to build the Redwood GDSR dataset. The LiDAR point cloud is registered to the RGB-D reference system, projected into each selected camera view and filtered to obtain the depth target and its validity mask. The consumer depth of the same frame is downsampled to obtain the low-resolution input, and the RGB image is used as guidance. 22

3.3 Comparison between the LiDAR reference and the consumer TSDF reconstruction for frame 300 of the *bedroom* scene. From left to right: LiDAR depth, TSDF depth, their difference, and depth difference as a function of distance. 24

3.4 Comparison between the consumer depth and the LiDAR-derived target. From left to right: consumer depth, projected LiDAR target, and pixel-wise depth error. 25

3.5 Target generation example. From left to right: consumer depth, LiDAR target after occlusion filtering, and densified LiDAR target. 26

3.6 Qualitative comparison on frame 100 of the *bedroom* test scene. From left to right: RGB guidance, degraded $\times 4$ input depth, DORNet prediction, FDSR prediction, and LiDAR reference. DORNet preserves more local structure in some regions but introduces visible artifacts, while FDSR produces a smoother reconstruction at the cost of attenuating fine geometric details. 32

5.1 Overview of BAF-Net. The depth branch (blue) sees only the sensor depth and produces the metric base D_0. The guidance branch (green) extracts features from the RGB image and from the monocular prior of Depth Anything V2, which is frozen (dashed). The features of the two branches are fused at four scales and a second decoder (orange) predicts the correction Δ that is added to D_0. 38

5.2 Building blocks of BAF-Net. (a) Residual block. (b) Encoder stage, which halves the resolution. (c) Fusion block, which combines depth and guidance features at the same resolution. (d) Decoder stage, which doubles the resolution and uses a skip connection. 41

5.3 Detailed architecture of BAF-Net. The two encoders E_D and E_G have the same structure but separate weights. Their features are fused at each of the four resolutions. The depth decoder uses only the features of E_D, so the metric base D_0 never depends on the RGB image. The guided decoder uses the fused features and, at full resolution, the features of a light stem. 42

6.1 Test RMSE on TOFDSR as a function of trainable parameter count, shown on a logarithmic horizontal scale. The dotted line indicates DORNet's RMSE. BAF-Net has the lowest RMSE among the plotted methods. 49

6.2 RMSE across the 560 test images. Left: cumulative distributions of per-image RMSE for the sensor input and BAF-Net. Right: input and BAF-Net RMSE for each image; points below the diagonal indicate an improvement over the input. 50

6.3 Validation RMSE during training for BAF-Net's final prediction $\hat{D}$, its intermediate metric base D_0, and the independently trained depth-only model. The dashed line shows the sensor-input RMSE. The RGB-only ablation is not included in this plot. 52

6.4 Reconstruction stages for the test image at the median improvement. From left to right: upsampled sensor depth, metric base D_0, guidance correction (centimeters), final prediction, and reference. Each panel reports its RMSE. 54

6.5 Depth reconstructions for five images selected at different percentiles of BAF-Net's improvement, followed by the worst case in the last row (six images in total). From left to right: RGB, sensor input, sensor-aligned Depth Anything V2, BAF-Net, and reference depth. Each row uses a shared color scale. White pixels have no valid reference measurement, and the values in the panels indicate RMSE. 55

6.6 Signed depth error (prediction minus reference) for the sensor input, sensor-aligned Depth Anything V2, and BAF-Net. The examples correspond to the 90th percentile, the 50th percentile, and the worst case. Red indicates overestimated distance; blue indicates underestimated distance. 56

6.7 Depth comparison on five TOFDSR test images, selected at the 10th, 30th, 50th, 70th, and 90th percentiles of BAF-Net's improvement over the input. Columns: RGB, ToF input, DORNet, BAF-Net, and reference. Dashed boxes mark selected regions; values are per-image RMSE (cm). 57

6.8 Signed depth errors for the five images in Figure 6.7. Columns: RGB and errors of the ToF input, DORNet, and BAF-Net. Red indicates overestimated depth and blue underestimated depth (-10 to 10 cm). Values are per-image RMSE. 58

Chapter 1

Introduction

1.1 Depth Maps and Depth Sensing Technologies

1.1.1 Depth Maps

A depth map is a two-dimensional representation of the three-dimensional geometry of a scene. Unlike an RGB image, where each pixel contains color information, each pixel in a depth map indicates the estimated distance from the sensor to a visible surface.

Depending on the sensor, depth measurements can contain noise, missing values, low spatial resolution, or errors near object boundaries and difficult surfaces. These characteristics are particularly important for Depth Super-Resolution, since the reconstruction problem depends strongly on the degradation introduced by the acquisition device.

Object boundaries are especially important because they usually correspond to changes in scene geometry, such as object contours or surfaces located at different distances. Preserving these boundaries is therefore essential for maintaining the correct 3D structure of the scene.

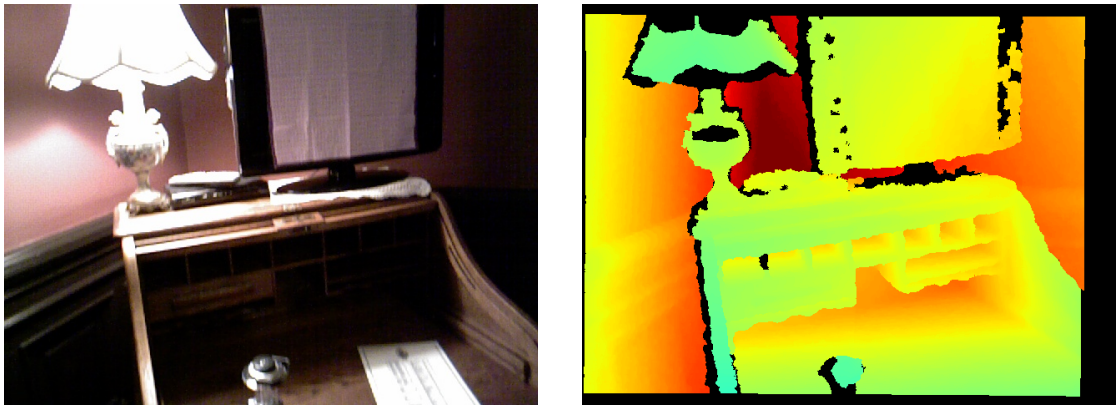


Figure 1.1: RGB image (left) and corresponding depth map (right). The depth map provides a pixel-wise representation of scene geometry, where each value is associated with the distance from the sensor to the observed surface.

1.1.2 Depth Acquisition Technologies

Depth information can be obtained using different types of sensors, each characterized by different accuracy, spatial resolution, operating range, and failure modes. The most relevant technologies for the purposes of this thesis include stereo vision, structured-light cameras, Time-of-Flight cameras and LiDAR systems.

Stereo vision. Stereo systems estimate depth from two or more images acquired from different viewpoints. Corresponding points are identified between the images and their disparity is converted into depth through camera geometry [1]. As a result, stereo techniques are passive and do not need any illumination. At the same time, stereo techniques' success depends on the presence of distinctive textures and can fail in conditions of poor illumination.

Structured light. Structured-light sensors project a known pattern in infrared onto the scene and capture the deformations of that pattern using the camera. Displacement of this projected pattern gives the disparity needed to measure depth. Examples of RGB-D sensors include Microsoft Kinect and Asus Xtion. They produce dense depth maps at relatively lower cost; however, their accuracy diminishes at larger distances and their data can be unreliable in certain cases, namely, near the edges of the objects, in reflections, and in cases where the projected pattern is not visible properly [2].

Time-of-Flight cameras. Time-of-Flight cameras measure distances through active illumination of the scene by measuring the propagation delay of the reflected

light. Depending on the implementation, the distance can be recovered from the direct travel time of a light pulse or from the phase shift of light [3].

ToF cameras provide a depth measurement for each sensor pixel and can acquire complete depth maps at high frame rates. The small form factor of the device makes it fit for mobile and consumer electronic devices. However, its resolution is often inferior to that of conventional RGB cameras and it is subject to various limitations such as interference and saturation.[3].

LiDAR. LiDAR (*Light Detection and Ranging*) systems also rely on active optical ranging, but they typically acquire individual distance measurements by scanning laser beams across the environment. As such, the output of the LIDAR system is essentially a three-dimensional point cloud as opposed to a dense depth map.

The use of LiDAR systems is extremely advantageous because they enable one to obtain very accurate and geometrically consistent measurements. However, the spatial sampling pattern of a LiDAR scan is fundamentally different from that of an RGB-D camera. In order to obtain a depth map from LiDAR data, the three-dimensional point cloud must be projected into a camera view using known calibration and camera poses.

1.1.3 RGB-D Cameras and LiDAR

RGB-D cameras provide depth maps from the camera's viewpoint. When the color and depth sensors are properly calibrated and their images are aligned, each depth measurement can be associated with a corresponding RGB pixel. This makes it straightforward to use RGB images as guidance for depth super-resolution.

LiDAR systems instead measure points in 3D space. High-precision LiDAR scanners can provide more accurate geometric measurements than consumer RGB-D cameras. However, projecting a LiDAR point cloud onto a camera image often produces a sparse depth map. Some regions may have no valid measurements because of differences in sensor viewpoints, occlusions, or incomplete scan coverage.

These differences must be considered when combining data from different sensors, especially when LiDAR measurements are used as reference depth for an RGB-D camera.

1.1.4 Applications of Depth Maps

Depth information plays an essential role in many computer vision applications, as it offers explicit information about the three-dimensional nature of the observed environment.

In 3D reconstruction and geometric modeling, depth measurements from different viewpoints can be combined to recover the geometry of a scene [4]. Depth is also used in augmented and virtual reality, where errors in the reconstructed geometry can affect how virtual objects appear and interact with the surrounding environment [5].

In robotics and scene understanding, depth provides information about object positions, distances, free space, and the spatial relationships between objects. Combined with visual information, these measurements help robotic systems navigate and interact with their environment [6].

Depth is also useful in tasks such as semantic segmentation and salient object detection. In RGB-D semantic segmentation, geometric information helps distinguish objects and regions that have a similar visual appearance [7, 8]. Similarly, in RGB-D salient object detection, depth can help separate foreground objects from the background when color information alone is insufficient [9].

However, the quality of the depth map obtained impacts the performance of these algorithms. Low resolution, noise, missing measurements due to occlusions, and inaccurate object boundaries can introduce errors in the final results. Improving the quality of depth maps can therefore benefit these applications.

1.2 Depth Super-Resolution

Depth Super-Resolution (DSR) aims to reconstruct a high-resolution depth map from a low-resolution input. This process can be expressed as

$$\hat{D}_{HR} = f(D_{LR}), \tag{1.1}$$

where D_{LR} is the input low-resolution depth map, $\hat{D}_{HR}$ is the reconstructed high-resolution depth map, and f is the reconstruction method. The reconstructed depth should preserve the overall geometry of the scene while recovering fine details and sharp depth boundaries.

The difference in resolution is described by the *scale factor*, denoted as $\times s$. Common scale factors in the literature are $\times 4$, $\times 8$, and $\times 16$. For example, with a scale factor of $\times 4$, each dimension of the depth map is increased by four, so each low-resolution pixel corresponds to an area of 4×4 pixels in the high-resolution image. Higher scale factors make reconstruction more difficult because more spatial information is missing from the input.

DSR is an underdetermined problem, since different high-resolution depth maps can produce the same low-resolution observation. Information lost during downsampling

or sensor acquisition cannot be recovered directly. Learning-based methods therefore use statistical and structural information learned from training data to estimate the missing details [10].

Most DSR methods have traditionally been evaluated using synthetic data. In this setting, low-resolution depth maps are generated by downsampling high-quality depth maps, usually with bicubic interpolation. This makes the degradation known and identical for every image, but it does not reproduce what happens with a real low-resolution sensor, where the measurements are also affected by noise, missing values and structural distortions [11, 12]. A model trained under synthetic degradation may therefore behave differently on real sensor data.

This thesis focuses on the real-world setting. The differences between synthetic and real-world protocols are discussed in Section 2.3.

1.3 RGB-Guided Depth Super-Resolution

A common approach to Depth Super-Resolution is to use an aligned high-resolution RGB image to provide additional information during reconstruction. This approach is known as RGB-guided Depth Super Resolution, or Guided Depth Super-Resolution (GDSR).

Given an RGB image I_{RGB} aligned with the depth observation, the reconstruction problem can be expressed as

$$\hat{D}_{HR} = f(D_{LR}, I_{RGB}). \tag{1.2}$$

The motivation behind GDSR is that high-resolution RGB images contain fine spatial information that is generally unavailable in the low-resolution depth measurement. In particular, object contours and thin structures can provide useful cues for reconstructing depth discontinuities.

Early methods such as MSG-Net showed that RGB guidance can improve the reconstruction of depth boundaries [10]. Later approaches introduced different ways to combine RGB and depth information, including adaptive filtering, frequency decomposition, and structure-aware feature fusion [13, 14, 15].

One limitation of RGB guidance is that RGB edges do not always correspond to changes in depth. Textures, shadows, and lighting variations can produce visible boundaries without any change in scene geometry. Transferring these structures to the reconstructed depth map can introduce incorrect details, so the network must learn which RGB information is useful for reconstruction [16].

Real sensor measurements introduce further difficulties. Input depth may contain noise, missing values, and structural distortions that simple downsampling cannot reproduce. Methods such as DORNet have investigated how to account for these degradations during reconstruction [12].

More recent approaches have also explored monocular depth foundation models as an additional source of geometric information [17]. Models such as Depth Anything V2 can estimate detailed scene structures from RGB images, although their predictions do not necessarily provide accurate metric distances [18].

1.4 Thesis Motivation and Objectives

The main objective of this thesis is to investigate Guided Depth Super-Resolution under realistic acquisition conditions and to design a neural architecture that will be able to combine metric measurements, RGB guidance and geometric priors effectively.

The work is organized into two main phases.

The first phase investigates the Redwood Indoor LiDAR-RGBD dataset [19] as a possible source of supervised real-world GDSR training data. This stage includes the development of a LiDAR-based target generation pipeline, geometric and photometric alignment analysis, and the evaluation of existing depth super-resolution architectures.

The objective of this preliminary study is not only to evaluate existing GDSR models on Redwood, but also to determine whether heterogeneous measurements acquired using a consumer RGB-D sensor and a professional LiDAR scanner can provide sufficiently consistent supervision for real-world GDSR.

The second phase focuses on the TOFDSR dataset [20], where the low-resolution input and high-quality reference are obtained through a dedicated real acquisition protocol. Within this setting, this thesis proposes BAF-Net, a new guided depth super-resolution architecture designed to combine metric depth reconstruction with high-resolution structural guidance.

The proposed model uses the low-resolution depth map as the metric reference, while RGB and monocular depth (produced with the help of Depth Anything V2 [18]) are used as additional sources of information. The model incorporates a multi-scale encoder–decoder structure enhanced by feature fusion to impose the geometric prior on the resulting depth map.

The goal is to find out whether the combination of the prior-guided approach and a real dataset can result in better depth reconstruction while minimizing the

unwanted influence of structures from the RGB data.

1.5 Thesis Organization

The remainder of this thesis is organized as follows.

Chapter 2 introduces the main concepts related to Depth Super-Resolution and Guided Depth Super-Resolution and reviews the most relevant approaches in the literature, with particular attention to the distinction between synthetic and real-world evaluation protocols.

Chapter 3 presents the evaluation conducted on the Redwood Indoor LiDAR-RGBD dataset [19], including the target generation pipeline, the evaluation of existing GDSR methods, and the analysis of the limitations identified during the experiments.

Chapter 4 describes the real-world experimental protocol adopted for TOFDSR [20], including data preprocessing, evaluation metrics, dataset splits, and training methodology.

Chapter 5 presents the proposed BAF-Net architecture and discusses its main components, including the metric depth branch, the foundation-model prior, the bilateral affine field, and the selective fusion mechanism.

Chapter 6 reports the experimental results and ablation studies and compares the different BAF-Net configurations with existing real-world Depth Super-Resolution methods.

Chapter **??** discusses the main findings, limitations, and implications of the experimental results.

Finally, Chapter **??** summarizes the contributions of this thesis and outlines possible directions for future work.

Chapter 2

Background and Related Work

This chapter introduces the main deep learning concepts used in Guided Depth Super-Resolution and reviews the most relevant approaches in the literature. Particular attention is given to RGB-D feature fusion, synthetic and real-world degradation protocols, real-world depth restoration, and the recent use of foundation models as geometric priors.

2.1 Deep Learning for GDSR

Most modern GDSR methods, including the architecture proposed in this thesis, rely on deep neural networks to reconstruct high-resolution depth maps from low-resolution measurements and RGB guidance. This section briefly introduces the main deep learning concepts used in these methods and in the proposed architecture. A more general overview of deep learning can be found in [21].

2.1.1 Supervised Learning

A neural network is a function whose behavior is controlled by a set of parameters, also called weights. In supervised learning, these parameters are learned from examples rather than defined manually. Each example consists of an input and a corresponding reference, called *ground truth*. In GDSR, the input consists of a low-resolution depth map D_{LR} and an RGB image I_{RGB}, while the reference is the corresponding high-resolution depth map D_{GT}.

During training, the network produces a prediction $\hat{D}$, and a *loss function* measures

the difference between this prediction and the reference. The network parameters are then updated to reduce the loss, using an optimization algorithm such as Adam [22]. This process is repeated over the training data, where one complete pass through the dataset is called an *epoch*.

The available data are usually divided into three parts. The *training set* is used to learn the parameters. The *validation set* is used during development, for example to choose which version of the model to keep. The *test set* is used only at the end, to measure the performance on images that the network has never seen. Keeping these sets separate is important: a network can obtain a very low error on the training images without working well on new ones, a problem known as *overfitting*.

2.1.2 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are widely used for image processing tasks, including depth reconstruction. Their basic operation is the convolution, where a small filter, for example of 3×3 pixels, is applied across an image to extract local information. The values of the filter are learned during training.

This operation has two useful properties. First, the same filter is applied at every position, so the number of parameters does not depend on the size of the image. Second, each output value depends only on a small neighborhood of the input, which fits the fact that nearby pixels are usually related.

A convolutional layer applies many filters in parallel. Its output is a set of *feature maps*, each one describing where a certain pattern appears in the image. By stacking several layers, with a non-linear function between them, the network can describe more and more complex structures: the first layers respond to simple patterns such as edges, while deeper layers respond to larger structures such as object parts.

In GDSR, convolutional layers are used to extract information from both depth and RGB images. These features can then be combined to recover geometric details that are missing or degraded in the low-resolution depth.

2.1.3 Encoder–Decoder Architectures

A common architecture for image reconstruction tasks is the encoder–decoder. In many GDSR methods, the low-resolution depth is first resized to the target resolution, and the network learns to refine it using the available guidance information.

The *encoder* progressively reduces the spatial resolution of the feature maps and increases their number. In this way each feature summarizes a larger region of the image, and the network can use information about the general structure of the

scene. The *decoder* does the opposite: it progressively increases the resolution until the original size is reached.

Reducing the resolution, however, loses fine details such as the exact position of the edges. To avoid this, *skip connections* pass the feature maps of the encoder directly to the decoder at the same resolution. This design was made popular by U-Net [23]. Since the features are processed at several resolutions, these architectures are also called *multi-scale*.

Encoder–decoder architectures are particularly suitable for GDSR because they allow the network to combine information about the overall scene geometry with local details needed to reconstruct object boundaries.

2.1.4 Residual Learning

Residual learning is a widely used technique for improving the training of deep neural networks. Instead of learning a complete transformation directly, a *residual block* learns a correction to its input, which is then added back to the original features. This approach was introduced in ResNet and helps the optimization of deep architectures [24].

A similar idea can be applied to depth super-resolution. The low-resolution depth, after being resized to the target resolution, already provides an initial estimate of the scene geometry, although it may contain noise and other errors. Rather than predicting the entire depth map from scratch, the network can learn a correction that is added to this initial estimate.

This principle is also used in the proposed architecture, where the final depth is obtained by refining an initial metric depth reconstruction.

2.1.5 Attention and Vision Transformers

Convolutional layers primarily extract information from local neighborhoods, so several layers are generally needed to relate distant regions of an image. The *attention* mechanism offers an alternative by allowing a network to model relationships between features at different spatial positions [25].

Vision Transformers (ViT) apply attention mechanisms to image representations. Typically, an image is divided into small patches, which are processed to learn relationships between different regions [26]. These architectures can capture both local and global information and are widely used in modern computer vision models.

Vision Transformers are also used in foundation models such as Depth Anything V2, which is described in Section 2.5. Such models are trained on large datasets and

can provide useful geometric information even for images that were not included in their training data.

2.1.6 Loss Functions

Learning-based methods are trained by minimizing a loss function, which measures the difference between the predicted depth $\hat{D}$ and the reference depth D_{GT}. Two common choices are the L1 and L2 losses:

$$\mathcal{L}_1 = \frac{1}{|\Omega|} \sum_{p \in \Omega} \left| \hat{D}(p) - D_{GT}(p) \right|, \qquad \mathcal{L}_2 = \frac{1}{|\Omega|} \sum_{p \in \Omega} \left(\hat{D}(p) - D_{GT}(p) \right)^2, \tag{2.1}$$

where Ω is the set of pixels with a valid reference value. Pixels without a valid reference are excluded, because real datasets often contain holes.

The L1 loss is less sensitive to large errors than L2, which penalizes larger differences more strongly. Another commonly used measure is the Root Mean Squared Error (RMSE), corresponding to the square root of the L2 loss. RMSE is the main evaluation metric adopted in this thesis and is also used as the main reconstruction loss of the proposed network.

However, minimizing these losses does not necessarily guarantee the reconstruction of sharp geometric details. Depth super-resolution is an ambiguous problem, since different high-resolution depth maps may be compatible with the same low-resolution observation. When the position of a depth boundary is uncertain, minimizing the reconstruction error can lead to overly smooth predictions [27].

To address this limitation, some methods introduce additional terms into the loss function. Examples include losses computed on depth gradients [15], losses in the frequency domain, and adversarial losses. These terms can encourage the reconstruction of sharper details, although they may also introduce structures that do not correspond to the actual scene geometry.

2.2 Approaches to Guided Depth Super-Resolution

Different families of algorithms have been developed to solve Guided Depth Super-Resolution problem over time. The existing families of algorithms can be categorized as filtering-based, optimization-based and learning-based methods [28]. While these families differ significantly in formulation, the common goal remains the same: exploiting information from a high-resolution guidance image while preserving the geometry contained in the depth measurements.

2.2.1 Filtering-Based Methods

Filtering-based approaches are one of the oldest techniques used in Guided Depth Super-Resolution. The core concept behind these techniques is the reconstruction of each depth pixel based on a weighted combination of neighboring pixels. The weight can be determined by spatial closeness and information from the high resolution guidance image.

A representative example is Joint Bilateral Upsampling (JBU) [29], which extends bilateral filtering by using the high-resolution RGB image to guide the interpolation of a low-resolution signal. The pixels having similar colours are given more weight on the assumption that RGB discontinuity implies geometric discontinuity.

The Guided Image Filtering (GF) algorithm uses an alternative formulation, in which the output within the local window is considered a linear transformation of the guidance image [30]. Other methods have proposed more advanced techniques to minimize the impact of inconsistent structures between the RGB and depth images. Static and Dynamic Filtering methods, for instance, use information from the guidance image together with its increasingly regularized versions to reduce the impact of misleading structures [31].

The main limitation of these methods is that the correspondence between RGB and depth edges is only approximate. Changes in texture, illumination, or material may produce strong RGB boundaries without any corresponding change in geometry, causing *texture-copying* artefacts [28].

More recent filtering approaches incorporate learning into the filtering process. DKN, for example, uses neural networks to predict spatially varying kernel weights and sampling locations from RGB and depth information [13]. The filtering principle is therefore preserved, while the filter itself becomes adaptive and learned from data.

2.2.2 Optimization-Based Methods

A second family formulates GDSR as an optimization problem. Rather than directly defining an interpolation rule, these methods search for a high-resolution depth map that remains consistent with the measured low-resolution depth while satisfying additional assumptions about the expected scene geometry.

A generic formulation can be written as

$$\hat{D}_{HR} = \arg\min_{D} \mathcal{L}_{data}(D, D_{LR}) + \lambda \mathcal{R}(D, I_{RGB}), \tag{2.2}$$

where $\mathcal{L}_{data}$ preserves consistency with the observed depth and $\mathcal{R}$ introduces a regularization prior, often using information from the RGB guidance image.

One of the early approaches modeled depth super-resolution through a Markov Random Field, where local depth consistency and RGB-dependent smoothness are combined in a global energy function [32]. Other methods exploit the piecewise-smooth structure of depth maps. Ferstl et al., for example, use anisotropic Total Generalized Variation to preserve sharp geometric discontinuities while regularizing smooth surfaces [33].

Low-rank and non-local priors have also been investigated. Lu et al. formulate depth enhancement as a low-rank matrix completion problem in order to reduce the dependence on potentially misleading RGB structures [34]. Graph-based approaches instead represent depth pixels as nodes of a weighted graph and exploit graph smoothness to preserve local geometry. Liu et al., for instance, combine an internal graph-based smoothness prior with an external color-guided gradient constraint [35].

Optimization-based approaches provide an explicit interpretation of the reconstruction process and allow geometric assumptions to be directly encoded into the objective. However, their performance depends strongly on the selected priors, and iterative optimization can make them computationally expensive compared with feed-forward neural networks [28].

2.2.3 Learning-Based Methods

Learning-based methods replace manually designed filtering rules or regularization functions with representations learned from data. Instead of explicitly defining how the RGB image should influence the depth reconstruction, this relationship is estimated from training examples.

Before the widespread adoption of deep neural networks, several approaches relied on *dictionary learning*. These methods represent RGB and depth patches as sparse combinations of learned basis elements. Tosic and Drewes, for example, learn coupled sparse representations of intensity and depth images [36]. Song et al. later introduced coupled dictionaries specifically designed to model both shared and modality-specific information between RGB and depth [37]. Convolutional dictionary learning further extends this idea by replacing patch-wise dictionaries with spatially invariant convolutional filters [38].

More recent GDSR approaches are predominantly based on *deep learning*. Rather than relying on a predefined reconstruction model, the network learns a non-linear mapping between the degraded depth, the guidance image, and the desired high-resolution depth.

A common GDSR architecture contains two feature extraction paths: one processes the depth observation and the other processes the RGB guidance. Their intermediate representations are then combined through a fusion mechanism before reconstructing the final depth map [28]. The main difference between modern architectures often lies precisely in how this cross-modal fusion is performed.

2.2.4 Deep Learning Architectures for GDSR

One of the first influential deep learning approaches was MSG-Net, which introduced multi-scale RGB guidance to progressively recover depth structures at different resolutions [10]. Similar two-stream architectures were later explored by Deep Joint Image Filtering, where separate convolutional branches extract features from the target and guidance images before combining them for reconstruction [39].

As the field developed, greater attention was placed on how guidance information should be transferred. DKN predicts deformable sampling locations and spatially varying filtering kernels instead of directly concatenating RGB and depth features [13]. FDSR separates guidance information into different frequency components through octave convolutions and mainly transfers high-frequency RGB information to the depth reconstruction branch [11].

Other approaches explicitly incorporate prior knowledge into the architecture. GraphSR combines a learned affinity graph with a differentiable optimization layer, merging data-driven feature extraction with graph-based regularization [40]. DCTNet similarly starts from an optimization formulation based on the Discrete Cosine Transform and incorporates it into a neural architecture. It also separates shared and modality-specific RGB-D features to reduce texture over-transfer [14].

Several methods instead focus on structural information. DADA interprets the reconstruction process as learned anisotropic diffusion, where information propagation is guided by image structure [16]. SGNet combines gradient and frequency representations to emphasize high-frequency geometric structures and improve the reconstruction of depth boundaries [15].

Transformer-based approaches provide another strategy for modeling RGB-depth relationships. By exploiting attention mechanisms, these architectures can capture long-range interactions between guidance and depth features, complementing the local information extracted by convolutional networks [41].

Despite these advances, the central difficulty remains unchanged: the RGB image contains significantly more high-frequency information than the depth map, but not all of it corresponds to geometry. Modern architectures therefore increasingly rely on attention, correlation, frequency decomposition, or selective fusion to control which guidance structures are transferred to the reconstructed depth.

More recent research has also focused on handling real sensor degradation and incorporating geometric priors from monocular depth foundation models. Approaches such as DORNet [12], DuCos [17], and WAVE [42] explore these directions through different reconstruction and fusion strategies. These developments are discussed in more detail in Sections 2.4 and 2.5.

2.3 Synthetic and Real-World DSR Protocols

A fundamental distinction in DSR research concerns how the low-resolution depth input is obtained.

In the conventional synthetic setting, the LR depth is generated directly from the high-resolution reference:

$$D_{LR}^{syn} = \mathcal{G}(D_{GT}), \tag{2.3}$$

where $\mathcal{G}$ is a predefined degradation operator, typically bicubic downsampling. The input and target therefore originate from the same depth map and are perfectly aligned.

This method has become quite popular due to its controllability and replicable training pairs. The recent research still uses the formulation to compare their models on a large scale. For instance, the authors of DuCos train on synthesized Hypersim data with bicubic interpolated LR input images [17]. Moreover, WAVE tests two popular GDSR protocols trained on Hypersim and NYU-v2 data and tested on multiple datasets and upscaling factors [42].

It is important to distinguish between the *origin of a dataset* and the *evaluation protocol*. A dataset containing real RGB-D acquisitions can still be used for a synthetic DSR protocol if its ground-truth depth is artificially downsampled to create the LR input.

Real-world DSR instead uses depth information from the measurement provided by an actual sensor for the LR input. The degradation process can then include sensor noise, missing measurements, blur and structural distortions.

FDSR introduced RGB-D-D as one of the first dedicated benchmarks for this setting, using real measurements rather than generating the LR input from the reference depth [11]. DORNet further highlights the difference between the two protocols, showing that real depth measurements present substantially stronger structural and statistical deviations from the ground truth than bicubically downsampled inputs [12].

Consequently, results obtained using synthetic LR depth can't be compared with results obtained using the actual LR measurements of the sensor, even when the same dataset is used for evaluation.

The two protocols described in this chapter are compared in Fig. 2.1.

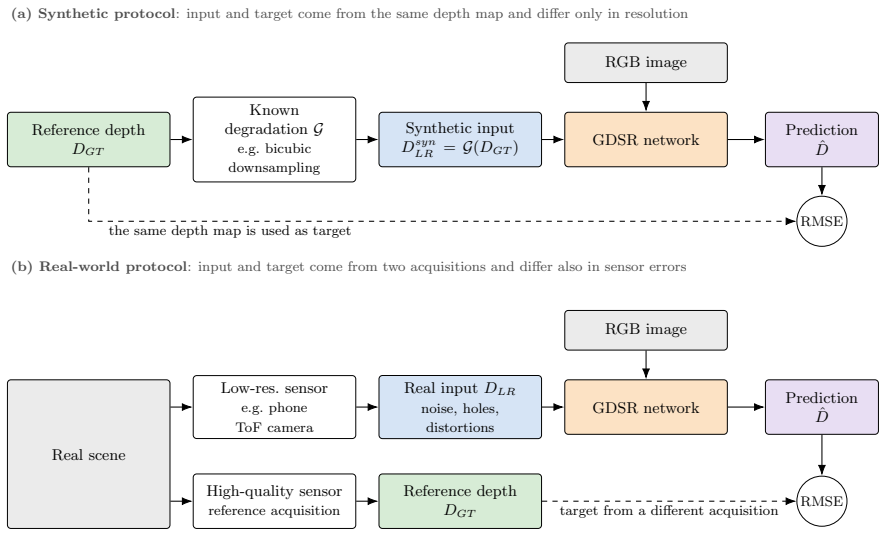


Figure 2.1: Synthetic and real-world DSR protocols. (a) In the synthetic protocol, the low-resolution input is generated from the reference depth with a known degradation, so input and target differ only in resolution. (b) In the real-world protocol, input and reference are acquired by two different sensors, so the input also contains the errors of the low-resolution sensor.

2.4 Real-World Guided Depth Super-Resolution

The increasing interest in real sensor degradation has motivated GDSR architectures specifically designed for unknown degradation.

FDSR introduced the RGB-D-D dataset and provided a baseline for evaluating depth super-resolution using real sensor measurements [11]. SFG later proposed a structure-flow mechanism to control the transfer of structural information between RGB and depth features [43].

DORNet focuses on the degradation of the input depth [12]. The network learns degradation representations from low-resolution measurements through a self-supervised strategy. Degradation-Oriented Feature Transformation modules then use these representations to adjust the RGB features before fusing them with the

depth information. In this way, the network can control the contribution of RGB guidance according to the degradation present in the input depth.

Unlike the synthetic setting, where the low-resolution input is generated through a known downsampling operation, real-world GDSR must also account for the errors introduced during sensor acquisition.

2.5 Depth Foundation Models as Geometric Priors

Monocular depth estimation models predict depth maps from a single RGB image. These models can recover detailed scene structures and object boundaries, even in images that were not part of their training data.

Relative and metric depth. Estimating depth from a single image is an ambiguous problem. For example, a small object close to the camera can produce a similar image to a larger object farther away. Many general-purpose monocular models therefore predict *relative* depth instead of absolute distances in meters. These predictions describe the geometry of a scene, but their scale and shift are generally unknown.

MiDaS [44] uses a scale- and shift-invariant loss in inverse depth space, allowing the model to learn from multiple datasets with different depth representations. DPT [45] later introduced a Vision Transformer backbone to improve the quality of depth predictions. Depth Anything [46] uses a DINOv2 encoder [47] and large amounts of unlabelled image data for training. Depth Anything V2 [18] further improves the reconstruction of object boundaries through synthetic supervision and pseudo-labelled real images.

Sensor fusion and metric alignment. Physical depth sensors provide measurements in metric units, but their depth maps may have limited resolution, noise, or missing values. Monocular models, on the other hand, produce dense depth predictions with detailed geometric structures, although their absolute distances may be inaccurate. Combining these two sources can help recover details while preserving the metric information provided by the sensor.

If R is the relative inverse depth predicted by a monocular model, its relationship with the metric depth D can be approximated as

$$\frac{1}{D} \approx \alpha R + \beta, \tag{2.4}$$

where α and β are the unknown scale and shift parameters, which can vary between images. When metric depth measurements are available, these parameters can be estimated using a least-squares fit. However, a single scale and shift for the entire image may not be sufficient to correct local errors in the monocular prediction. Learned fusion methods can address this limitation by combining sensor measurements with monocular depth information. Prior Depth Anything [48], for example, combines metric alignment with learned refinement to improve depth estimation.

Foundation priors in GDSR. Depth foundation models have also been used in Guided Depth Super-Resolution. DuCos [17] uses features and relative depth predictions from a frozen depth foundation model as additional guidance. It combines these features with the measured depth through correlation-based fusion and uses gradient information to improve the reconstruction of geometric details.

WAVE [42] uses high-level image features extracted by DINO together with separately processed color information. The network first reconstructs the coarse structure of the scene and then progressively introduces finer details from the RGB image.

In both approaches, foundation models provide structural or semantic information that can support depth reconstruction without requiring their predictions to be accurate in metric units.

2.6 Common GDSR Benchmarks

Several datasets are commonly adopted in DSR and GDSR research.

NYU Depth V2. NYU Depth V2 contains RGB-D acquisitions from indoor environments captured using a Microsoft Kinect [49]. It is one of the most widely used benchmarks for synthetic GDSR, where LR depth is typically generated by downsampling the available reference depth.

Middlebury. The Middlebury stereo datasets provide high-accuracy disparity and depth references obtained using structured-light-assisted acquisition [50]. They are frequently used for cross-dataset evaluation of GDSR methods.

Lu. The dataset introduced by Lu et al. contains RGB-D scenes originally used for depth enhancement and restoration [34]. It is also commonly adopted as an external test set in GDSR benchmarks.

RGB-D-D. RGB-D-D was introduced specifically for real-world depth super-resolution and contains depth observations acquired from real sensing systems [11]. It can be used either with its genuine LR measurements for real-world evaluation or with synthetically generated LR inputs in conventional benchmarking protocols.

TOFDSR. TOFDSR originates from real Time-of-Flight acquisitions. The underlying TOFDC data were captured using the ToF sensor and RGB camera of a Huawei P30 Pro, with reference depth obtained using a Helios ToF camera [51]. DORNet constructs the TOFDSR setting from these data, using approximately 10,000 training pairs and 560 test pairs for real-world DSR evaluation [12].

DIML. The DIML/CVL dataset contains a large collection of RGB-D images from indoor and outdoor environments [52]. It is increasingly used as an additional cross-dataset benchmark in recent GDSR works such as WAVE.

Hypersim. Hypersim is a photorealistic synthetic indoor dataset containing 77,400 rendered images from 461 scenes with dense geometric annotations [53]. Recent methods such as DuCos and WAVE use subsets of Hypersim as synthetic GDSR training data to evaluate cross-dataset generalization.

Chapter 3

Evaluation of Redwood Indoor LiDAR-RGBD

The first part of this thesis examines whether the Redwood Indoor LiDAR-RGBD dataset can be used to train GDSR models on real sensor data. Redwood contains RGB-D sequences captured with a consumer camera and detailed 3D reconstructions obtained through terrestrial laser scanning [19]. The two sources make it possible to compare consumer depth measurements with a separate geometric reference.

Redwood was not collected for depth super-resolution, so it does not contain ready-to-use pairs of low- and high-resolution depth maps. To obtain reference maps, the LiDAR geometry must first be aligned with the RGB-D camera trajectory and then projected into the individual camera views.

This chapter describes how the training pairs were generated, how FDSR and DORNet performed on them, and which limitations were observed. The results also explain why the following experiments use the TOFDSR protocol.

3.1 Dataset and Experimental Formulation

Redwood contains five indoor scenes: *apartment*, *bedroom*, *boardroom*, *lobby*, and *loft* [19]. The RGB-D sequences were recorded with an Asus Xtion Live at 640×480 pixels and approximately 30 frames per second. Depth is stored in millimeters and is registered to the corresponding RGB image. Figure 3.1 shows examples from each scene.

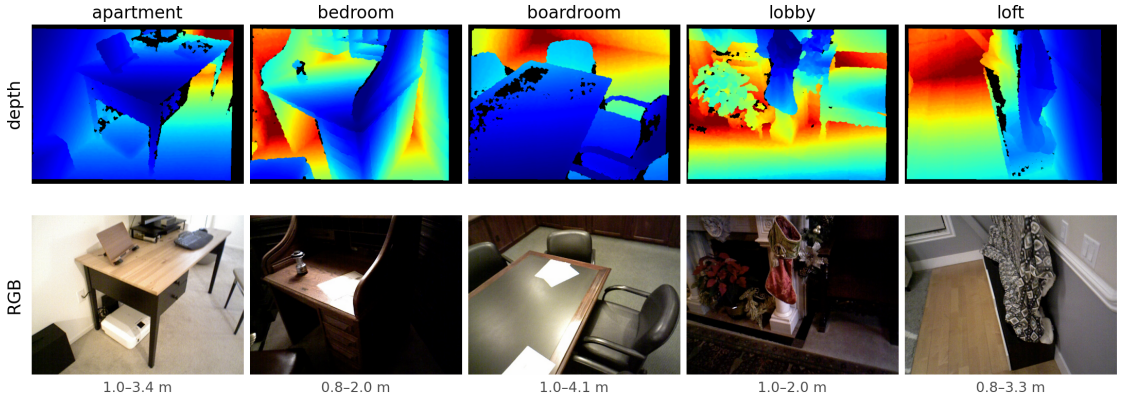


Figure 3.1: Examples from the five Redwood Indoor LiDAR-RGBD scenes used in this work. The top row shows depth maps acquired by the consumer RGB-D sensor, while the bottom row shows the corresponding RGB images. The depth ranges reported below each scene indicate the approximate interval visualized in each depth map.

The LiDAR reconstruction and the camera trajectory use different coordinate systems. After aligning the two systems, the recorded camera poses can be used to project the LiDAR point cloud into each image and obtain a reference depth map.

Each training sample contains the RGB guidance image, a degraded version of the consumer depth, and the depth target obtained from LiDAR. Validity masks are included for both depth maps. The sample is written as

$$(I_{RGB}, D_{LR}, M_{LR}, D_{GT}, M_{GT}), \tag{3.1}$$

Here I_{RGB} is the RGB image, D_{LR} is the degraded consumer depth, and M_{LR} marks valid consumer measurements. The LiDAR-derived target is D_{GT}, while M_{GT} identifies pixels supported by actual LiDAR observations. Figure 3.2 summarizes the pipeline used to build these samples; each step is described in the following sections.

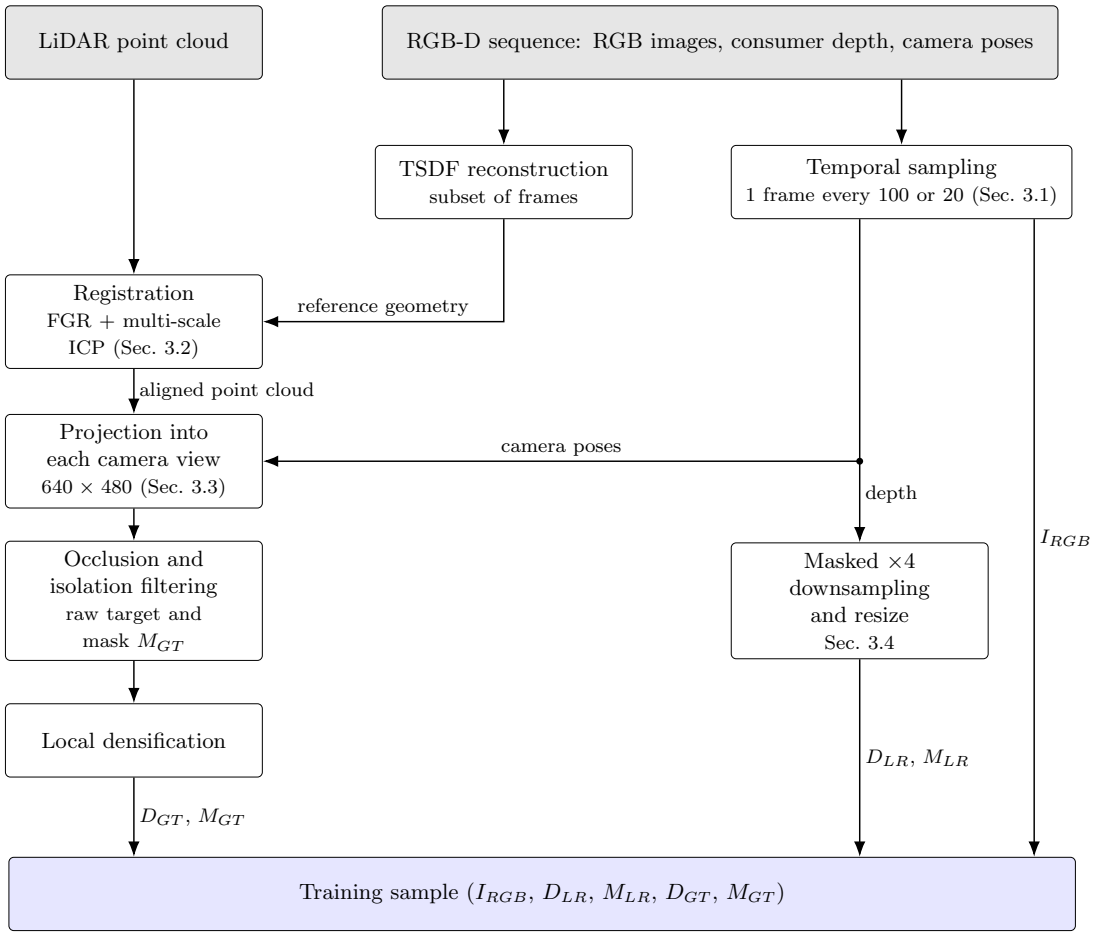


Figure 3.2: Overview of the pipeline used to build the Redwood GDSR dataset. The LiDAR point cloud is registered to the RGB-D reference system, projected into each selected camera view and filtered to obtain the depth target and its validity mask. The consumer depth of the same frame is downsampled to obtain the low-resolution input, and the RGB image is used as guidance.

3.2 LiDAR-to-Camera Registration

The LiDAR point cloud and the RGB-D camera trajectory are defined in different coordinate systems. Therefore, before generating the depth targets, the LiDAR geometry must be aligned with the RGB-D reference system by estimating a transformation for each scene.

This process is not a minor correction. In one scene, the coordinate systems were separated by approximately 53 m and differed in orientation by nearly 90°.

Projecting the unregistered LiDAR points into the camera images would therefore place the geometry in the wrong locations.

Registration was carried out in three stages: construction of an RGB-D reference, coarse global alignment, and local refinement.

First, a reference point cloud was reconstructed from a subset of the consumer depth frames using a Truncated Signed Distance Function (TSDF). The camera poses were already available, so the reconstructed geometry was expressed in the RGB-D coordinate system. A TSDF divides space into voxels and stores a signed distance to the nearest surface within a limited distance range [54, 55]. Distances are positive in front of the surface and negative behind it. Integrating multiple frames averages their measurements and reduces part of the sensor noise. The final surface is extracted where the signed distance is zero.

The LiDAR cloud was then aligned coarsely with this reference using Fast Global Registration (FGR) and Fast Point Feature Histograms (FPFH). FPFH describes the local geometry around each point through the relative orientations of nearby surface normals [56]. Because these descriptors do not depend on the global position or orientation of the point cloud, they can be used to find correspondences between initially misaligned reconstructions. FGR estimates a rigid transformation from these correspondences while reducing the influence of incorrect matches [57]. It can handle the large initial difference between the coordinate systems, although the estimated alignment still needs refinement.

The final stage uses multi-scale point-to-plane Iterative Closest Point (ICP). Standard ICP repeatedly finds corresponding points in two clouds and updates the transformation to reduce their distances [58]. In the point-to-plane formulation, the residual is measured along the surface normal of the matched point, which often improves convergence on locally smooth surfaces [59]. The FGR result provides the initial transformation. ICP is then run from coarser to finer voxel resolutions to refine the alignment without relying on fine-scale correspondences from the beginning.

The resulting registration affects the accuracy of every generated target. Even when the LiDAR reconstruction is accurate, errors in the RGB-D poses or in the transformation between the two systems can produce differences between the projected reference and the consumer measurements. Figure 3.3 compares the LiDAR and consumer TSDF depths for a frame from *bedroom*. Their overall geometry is similar, but the error map contains local differences, and the plot shows a small systematic variation with distance.

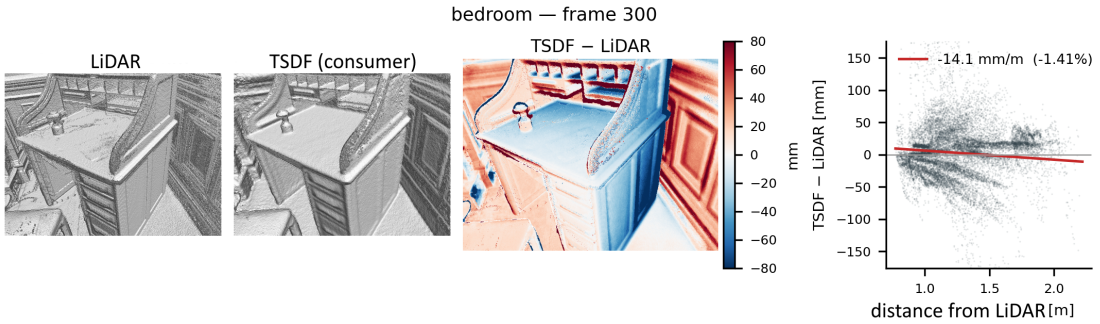


Figure 3.3: Comparison between the LiDAR reference and the consumer TSDF reconstruction for frame 300 of the *bedroom* scene. From left to right: LiDAR depth, TSDF depth, their difference, and depth difference as a function of distance.

3.3 LiDAR Target Generation

Once the LiDAR reconstruction has been aligned with the camera trajectory, the point cloud can be projected into each selected frame to create a depth target.

For a 3D point with camera coordinates (X, Y, Z), the image position is calculated with the pinhole camera model:

$$u = f_x \frac{X}{Z} + c_x, \qquad v = f_y \frac{Y}{Z} + c_y, \tag{3.2}$$

The camera intrinsics used for this projection are

$$f_x = f_y = 525, \qquad c_x = 319.5, \qquad c_y = 239.5. \tag{3.3}$$

The depth targets are rendered at the native resolution of 640×480 pixels. Only LiDAR points with depths in the range $[0.1{,}10]\,\mathrm{m}$ are used. Figure 3.4 compares an initial LiDAR projection with the consumer depth image and shows the corresponding pixel-wise errors.

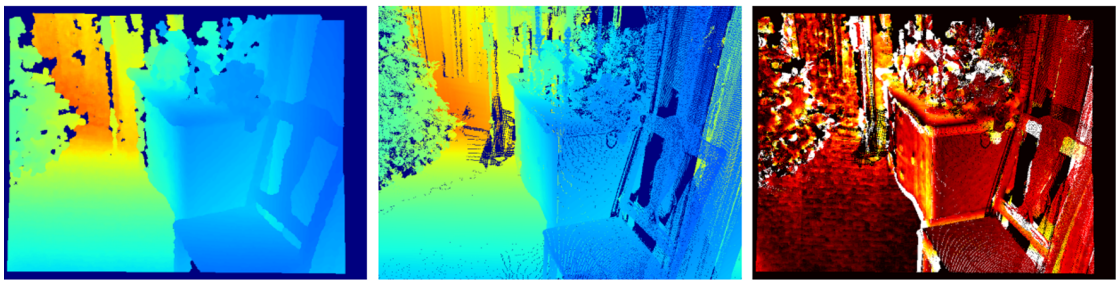


Figure 3.4: Comparison between the consumer depth and the LiDAR-derived target. From left to right: consumer depth, projected LiDAR target, and pixel-wise depth error.

3.3.1 Visibility and Occlusion Filtering

The LiDAR reconstruction is a point cloud, not a continuous surface. Direct projection with a z-buffer can leave gaps near object boundaries. Through these gaps, background points may appear in front of regions that should be occluded, producing false depth values.

The pipeline reduces these errors by checking each projected point against nearby measurements. Points that lie substantially farther away than the surrounding surface are removed because they may belong to a background region hidden by a foreground object. Another filter rejects isolated projections that lack nearby points at a similar depth.

The filtering thresholds were chosen to remove incorrect projections without discarding too much valid LiDAR data. The isolation test uses a 5×5 neighborhood and accepts a point only when enough neighboring measurements differ by no more than 0.05 m in depth. This mainly removes isolated points while retaining locally consistent measurements.

3.3.2 Raw Targets, Densification and Validity Masks

Two versions of the depth target are generated for each frame. The *raw LiDAR target* contains only pixels reached by projected LiDAR points. A second, densified version fills small gaps using nearby valid depths. This interpolation is local and does not extend across strong depth discontinuities.

The interpolated values are kept separate from measured reference data. A binary mask M_{GT} records which pixels were supported by LiDAR before densification:

$$M_{GT}(p) = \begin{cases} 1, & \text{if pixel } p \text{ is directly supported by LiDAR,} \\ 0, & \text{otherwise.} \end{cases} \tag{3.4}$$

The densified map is useful for visualization and for operations that benefit from a more continuous depth image. During supervised training, however, the loss is evaluated only at pixels marked as valid in M_{GT}. Figure 3.5 shows the consumer depth, the filtered LiDAR target, and the densified result.

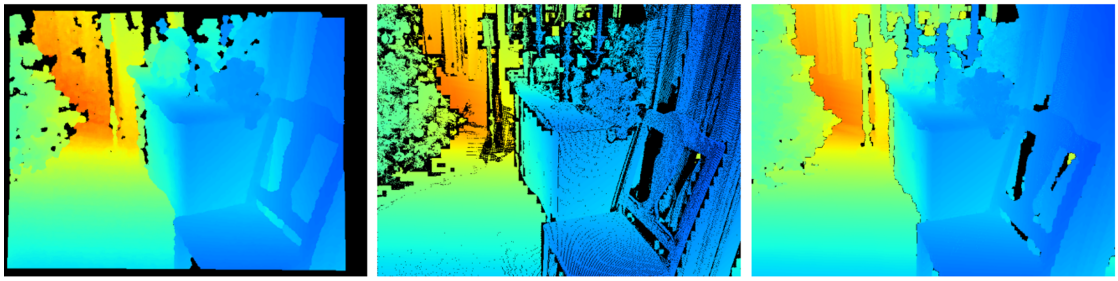


Figure 3.5: Target generation example. From left to right: consumer depth, LiDAR target after occlusion filtering, and densified LiDAR target.

3.3.3 Partial LiDAR Coverage

Many pixels have no LiDAR reference after projection and filtering. In the *bedroom* sequence, the mean valid coverage was approximately 56.5%, ranging from about 16.5% to 91.1% across individual frames. Mean coverage in the scenes used for the experiments was approximately 56–68%.

This leaves a substantial portion of a typical image without directly measured target depth. The missing regions depend on which surfaces were observed by the original laser scans and how those observations project into each camera view.

Increasing the rendering resolution cannot produce new LiDAR observations. The same projected points are distributed over a larger image, so fewer pixels receive a measurement. Table 3.1 reports the raw projection density and coverage at different resolutions, before occlusion filtering. Both decrease as the rendering resolution increases. The native 640×480 resolution was therefore retained for the target maps.

Table 3.1: Effect of the rendering resolution on the projected LiDAR points. Values are averaged over the five scenes, using 100 frames per scene. The raw coverage is the fraction of pixels that receive at least one LiDAR point, before the occlusion filtering.

Rendering resolution	pt/px	Raw coverage
640×480	2.90	72.3%
800×600	1.86	60.7%
1280×720	0.97	43.3%
1280×900	0.77	38.1%

Because the LiDAR reference is incomplete, training errors are measured only where the ground-truth mask is valid. For example, a masked L1 reconstruction loss is defined as

$$\mathcal{L}_{rec} = \frac{\sum_p M_{GT}(p) \left| \hat{D}(p) - D_{GT}(p) \right|}{\max\left(\sum_p M_{GT}(p), 1\right)}. \tag{3.5}$$

All Redwood experiments used masked supervision. Some configurations also included additional weighting or filtering terms.

3.4 Construction of the $\times 4$ Input

Both the consumer depth and the LiDAR target are represented at 640×480 pixels. Rendering the LiDAR point cloud at a higher resolution would reduce the proportion of valid target pixels without adding new measurements, as described in Section 3.3. For this reason, the reference resolution was kept unchanged.

To create a $\times 4$ super-resolution task, the original consumer depth was downsampled by a factor of four and then resized to the target image size before being passed to the network. The LiDAR target was not used to generate this low-resolution input.

This distinction matters when comparing the experiment with synthetic GDSR. In the synthetic protocol, the low-resolution map comes directly from the ground truth. In Redwood, the input comes from the consumer RGB-D camera and the reference comes from LiDAR. Their differences therefore include the loss of spatial resolution as well as disagreement between the two sensing systems.

Approximately 15% of consumer depth pixels can be invalid. Averaging a 4×4 block without accounting for them would treat missing measurements, often stored

as zeros, as valid depths. This can create large errors around holes and boundaries. The downsampling operation instead averages only valid values:

$$D_{LR}(q) = \frac{\sum_{p \in \mathcal{B}(q)} M(p) D(p)}{\max\left(\sum_{p \in \mathcal{B}(q)} M(p), 1\right)}, \tag{3.6}$$

Here $\mathcal{B}(q)$ denotes the 4×4 block associated with low-resolution pixel q, and M is the consumer validity mask. Invalid pixels do not contribute to the average.

3.5 Experimental Protocol

The primary test scene is *bedroom*; the other four scenes are used for training. This split is made by scene rather than by individual frame. Consecutive frames often show almost the same surfaces, so randomly assigning frames to training and testing could give the models access to nearly identical observations in both sets.

Each Redwood scene contains a continuous RGB-D sequence with many consecutive frames showing similar geometry. To reduce temporal redundancy, the initial experiment selects one frame every 100 frames, resulting in approximately 1,237 samples across the five scenes.

A second configuration uses a denser temporal sampling, selecting one frame every 20 frames. This increases the number of available samples by approximately a factor of five, although neighboring frames are more similar. The purpose of this experiment is to investigate whether the limited performance observed with the initial configuration is mainly related to the amount of training data.

In both configurations, *bedroom* remained the main held-out scene.

For cross-domain evaluation, pretrained networks were tested on *bedroom* without further training. In the Redwood-specific experiments, the networks were retrained on the remaining scenes. Within each configuration, all methods used the same degraded inputs and LiDAR validity masks.

The main metric was Root Mean Squared Error (RMSE), calculated only where a LiDAR reference was available:

$$\mathrm{RMSE} = \sqrt{\frac{1}{|M|} \sum_{p \in M} \left(\hat{D}(p) - D_{GT}(p)\right)^2}, \tag{3.7}$$

In this equation, M is the set of pixels with valid LiDAR measurements.

3.6 Baseline Networks

Two GDSR networks were used in the Redwood experiments: FDSR [11] and DORNet [12].

FDSR was chosen as a relatively lightweight baseline for real-world GDSR. It processes RGB information at different frequencies and transfers high-frequency guidance to the depth reconstruction. The network has approximately 0.6 million parameters.

DORNet was chosen because it is designed to handle unknown depth degradation. It learns a representation of the degradation from the input depth and uses that representation when combining RGB and depth features.

3.7 Cross-Domain Evaluation

The first test examined whether models trained on other datasets could be applied directly to Redwood without fine-tuning. This provides an initial measure of transfer to the consumer-depth and LiDAR-reference setting.

The FDSR and DORNet checkpoints trained on NYU Depth V2 were used to represent training with synthetic low-resolution inputs. A DORNet checkpoint trained on RGB-D-D was also tested to examine transfer from another dataset containing real sensor measurements.

All checkpoints were evaluated on the same 210 frames from *bedroom*, using identical input depths and masks. Table 3.2 reports the RMSE values.

Table 3.2: Cross-domain evaluation on the 210-frame *bedroom* test set. RMSE values are reported in millimeters.

Model	Training domain	RMSE
Input depth	–	170.8
FDSR	NYU-v2	173.5
DORNet	NYU-v2	173.8
DORNet	RGB-D-D	178.3

None of the pretrained models performed better than the degraded input, whose RMSE was 170.8 mm. The FDSR and DORNet checkpoints trained on NYU-v2 reached 173.5 mm and 173.8 mm, respectively. The RGB-D-D-trained DORNet checkpoint reached 178.3 mm. On these frames, training on another real-world depth dataset did not lead to better transfer than training on NYU-v2. The

results show that the tested pretrained models did not generalize directly to the consumer–LiDAR measurements of Redwood.

3.8 Retraining on Redwood

FDSR and DORNet were then trained on the Redwood pairs using masked supervision. The initial experiments used one frame every 100 frames, with *bedroom* held out for testing and the remaining scenes used for training.

Table 3.3 gives the results. The input RMSE was 170.8 mm. After Redwood-specific training, FDSR reached 150.9 mm, an improvement of 11.7%, and DORNet reached approximately 148.0 mm.

Table 3.3: Redwood-specific training results on the main *bedroom* test split using a temporal sampling step of 100. RMSE values are reported in millimeters.

Model	Training domain	RMSE
Input depth	–	170.8
FDSR	Redwood	150.9
DORNet	Redwood	**148.0**

Both models therefore improved when trained on the consumer–LiDAR pairs, unlike their pretrained versions. Their reductions in RMSE were nevertheless modest, at approximately 12–13% on the main *bedroom* test set.

3.8.1 Effect of Denser Temporal Sampling

Selecting one frame every 100 frames reduced temporal repetition but also limited the number of training examples. To test the effect of having more data, FDSR and DORNet were trained again with a sampling step of 20. This added approximately five times as many frames, including views that were often very similar to one another.

The main test environment was still *bedroom*. With the denser sampling, its test set increased from 210 to 1,056 frames. Table 3.4 compares the two experiments.

Table 3.4: Effect of the temporal sampling density on FDSR and DORNet for the main *bedroom* test scene. RMSE values are reported in millimeters.

Sampling step	Test frames	Input RMSE	FDSR RMSE	DORNet RMSE
100	210	170.8	150.9	148.0
20	1,056	169.6	160.2	146.6

FDSR improved over the input in both experiments, but its RMSE on the denser test set was 160.2 mm, compared with 150.9 mm in the original experiment. Relative to their respective input baselines, the reductions were 5.6% and 11.7%. Adding more frames did not yield a better result for FDSR in this comparison.

DORNet reached 146.6 mm with step 20, compared with 148.0 mm with step 100. Relative to the input RMSE of 169.6 mm for step 20, this is an improvement of approximately 13.6%. The change compared with the original result was small.

In the denser experiment, DORNet was trained for 40 epochs. The reported value of 146.6 mm was observed at epoch 20. Its error increased during later training and reached 173.6 mm at epoch 40, above the degraded-input RMSE. Performance on the held-out scene therefore became worse after the earlier checkpoint.

The two sampling settings also contain different test frames, so the reported RMSE values do not constitute a direct comparison on exactly the same images. The extra frames increase the number of observations, but much of the additional content is repeated scene geometry. In these experiments, the larger dataset did not produce consistent improvements across both networks.

These results suggest that selecting more frames from the same sequences is insufficient, by itself, to produce large gains on Redwood. FDSR showed a smaller relative improvement in the denser *bedroom* setting, while DORNet achieved its best reported result at epoch 20 and then deteriorated with continued training. The observations are consistent with temporal redundancy and the sensor differences examined below.

3.8.2 Qualitative Comparison

Figure 3.6 compares the two networks on a frame from the *bedroom* test sequence. DORNet retains more detail around the cabinet but produces artifacts on the chair and additional background structures that are not clearly present in the LiDAR reference. FDSR gives a smoother result. In particular, it smooths the chair geometry and loses some smaller features, including the holes in the backrest.

Both outputs remain visually closer to the degraded consumer depth than to the LiDAR target. The reference also has a different distribution of holes and sparsely observed areas, which does not match the missing measurements in the consumer image. Neither network reproduces these LiDAR-specific patterns.

The visual comparison illustrates a difficulty of this dataset: a prediction must account for differences between the two sensors as well as recover details lost during downsampling.

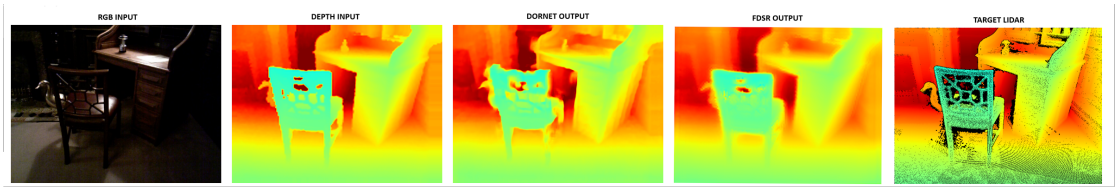


Figure 3.6: Qualitative comparison on frame 100 of the *bedroom* test scene. From left to right: RGB guidance, degraded ×4 input depth, DORNet prediction, FDSR prediction, and LiDAR reference. DORNet preserves more local structure in some regions but introduces visible artifacts, while FDSR produces a smoother reconstruction at the cost of attenuating fine geometric details.

3.9 Resolution Loss and Cross-Sensor Disagreement

The experiments were examined further by comparing the error already present in the original consumer depth with the additional error introduced by the artificial ×4 degradation.

On *bedroom*, the native consumer depth had an RMSE of approximately 159.2 mm on valid measurements. After downsampling by a factor of four and resizing to the network input resolution, the RMSE was 170.8 mm.

Much of the disagreement with the LiDAR reference was therefore present before the resolution was reduced. A network trained on these pairs must learn to correct differences between two sensing systems as well as reconstruct the spatial details removed by downsampling.

Several factors can contribute to this discrepancy, including sensor-specific biases, missing values, different responses to surface materials, and remaining registration errors. These effects are separate from the loss of spatial detail usually modeled in depth super-resolution.

The measured RMSE includes both sources of error. The comparable performance of FDSR and DORNet after retraining suggests that a more complex reconstruction network does not, by itself, resolve the uncertainty caused by the reference sensor.

The denser-sampling results provide further evidence. Increasing the number of frames by approximately five times did not produce a consistent increase in accuracy. FDSR improved the input RMSE by 5.6% on the main *bedroom* test set and by less than approximately 11% on each of the other scenes.

Redwood-specific training did reduce the error, but the improvements remained

limited. The high error of the native consumer depth relative to LiDAR, together with the results of denser sampling, suggests that dataset size alone was not the main obstacle in these experiments. Cross-sensor differences account for a substantial part of the evaluation error and are not directly related to the spatial resolution loss that GDSR is intended to address.

3.10 Limitations of Redwood as a GDSR Training Benchmark

The experiments identified several limitations of Redwood as a source of supervised GDSR training pairs.

Consecutive frames in each RGB-D sequence contain many of the same surfaces. The number of available images is therefore much larger than the number of distinct geometric observations, and increasing sampling density also increases redundancy.

The LiDAR target is incomplete. Depending on the scene and camera viewpoint, large parts of an image may have no valid reference value, so only a subset of the pixels can contribute to supervised training.

The LiDAR geometry must also be registered to the RGB-D trajectory. Any remaining errors in this alignment affect the projected targets. Invalid consumer depth measurements introduce a separate preprocessing problem and must be excluded when constructing the $\times 4$ input.

The main difficulty is the difference between the sensing technologies. The low-resolution input and the reference differ in their measurement errors and coverage as well as their spatial resolution. Redwood therefore combines depth super-resolution with the problem of matching observations from two different sensors. This makes its supervision less direct than that of datasets designed specifically for real-world GDSR.

Chapter 4

Real-World GDSR Experimental Protocol

4.1 Real-World GDSR Protocol

As discussed in Section 2.3, real-world GDSR uses low-resolution depth acquired by physical sensors rather than synthetically generated from the ground truth. This allows models to be evaluated under actual sensor degradation.

The preliminary Redwood study presented in Chapter 3 investigated this setting using consumer RGB-D measurements and LiDAR-derived references. However, generating these references required an additional registration and projection pipeline, introducing discrepancies between the input and target depth maps.

For the main experiments, this thesis therefore adopts the established real-world TOFDSR protocol [20]. The dataset provides paired low-resolution sensor measurements and reference depth maps, allowing the proposed architecture to be evaluated without the additional LiDAR target-generation procedure required for Redwood.

4.2 TOFDSR Dataset

TOFDSR is a real-world RGB-D dataset designed for depth super-resolution. Each sample contains an RGB image, a low-resolution depth measurement, and a corresponding reference depth map.

The dataset contains approximately 10 000 training samples and 560 test samples

[20, 12]. In this work, the $\times 4$ setting is considered throughout the main experiments.

A subset of 500 samples from the training set was reserved for validation. This validation set was used for model selection and for the architectural ablations, while the official test set was kept separate from model development.

4.3 Input and Ground-Truth Preparation

The TOFDSR low-resolution depth is directly obtained from the sensor. Therefore, no additional synthetic degradation is applied.

Since the LR depth has a lower spatial resolution than the reference, it is resized to the ground-truth resolution using bicubic interpolation before being provided to the network. Importantly, this operation only changes the spatial resolution of the input representation: the depth values still originate from the real sensor measurement.

All depth maps are converted to meters before being processed by the proposed model. The original depth values are stored in millimeters and are converted according to

$$D_{\mathrm{m}} = 10^{-3} D_{\mathrm{mm}}. \tag{4.1}$$

RGB images are represented using floating-point values in the range [0,1]. Separate validity masks are maintained for the input depth and the ground truth, since missing measurements in the LR depth do not necessarily correspond to invalid pixels in the reference.

4.4 Evaluation Protocol

To ensure comparability with previous works evaluated on TOFDSR, all models are tested using the same preprocessing and evaluation procedure.

Predictions are evaluated only at pixels where the reference depth lies in the valid range between 0.1 and 5.0 m. In addition, six pixels are removed from each image border before computing the error.

The main evaluation metric is the Root Mean Squared Error (RMSE):

$$\mathrm{RMSE} = \sqrt{\frac{1}{|M|} \sum_{p \in M} \left(\hat{D}(p) - D_{GT}(p) \right)^2}, \tag{4.2}$$

where M denotes the set of valid pixels after masking and border removal.

The RMSE is computed independently for each image and then averaged over the complete evaluation set. Results are reported in centimeters, following the convention commonly used for TOFDSR.

4.4.1 Validation and Model Selection

The validation subset is used for model selection and for all architectural ablations. The official test set is not used to select model variants or training checkpoints.

This separation ensures that architectural decisions are based only on the training and validation data, while the test set is reserved for the final evaluation of the selected model.

4.4.2 Protocol Verification

As an additional verification step, the adopted evaluation pipeline was compared with an independently released implementation from the literature.

DORNet [12] was selected for this purpose because its official model weights and TOFDSR evaluation code are publicly available. Using the released model, an RMSE of 4.212 cm was reproduced, consistent with the 4.21 cm reported by the authors.

The same predictions were then evaluated using the metric implementation developed for this thesis. The two implementations produced equivalent RMSE values up to numerical precision. The input depth also produced the same RMSE of 7.029 cm with both evaluation pipelines.

This experiment does not define the protocol itself, but provides an independent verification that the data preprocessing and evaluation implemented in this work are consistent with an existing real-world GDSR evaluation on TOFDSR.

Chapter 5

BAF-Net

This chapter presents BAF-Net (Boundary-Aware Fusion Network), the Guided Depth Super-Resolution network proposed in this thesis. The chapter first explains the idea behind the design, then describes the inputs, the architecture and its main properties, and finally the loss function and the training procedure. The results obtained with the model are reported in Chapter 6.

5.1 Design Overview

BAF-Net receives three inputs: the low-resolution depth map measured by the Time-of-Flight sensor, the high-resolution RGB image, and a monocular depth map estimated from the RGB image by Depth Anything V2 [18]. Each of them carries a different kind of information.

- The *sensor depth* is the only input expressed in meters. However, it has a low resolution and it is affected by noise, systematic errors and distortions near object borders.
- The *RGB image* has a high resolution and sharp edges, but many of its edges are due to texture, shadows or illumination and do not correspond to a change in depth.
- The *monocular prior* has sharp and clean object boundaries and does not contain RGB texture. However, as discussed in Section 2.5, it is a relative depth: it is correct only up to an unknown scale and shift, and it cannot be used directly as a metric measurement.

The architecture is built on a simple principle: *the metric values come from the sensor, the fine structure comes from the guidance.* For this reason the reconstruction

is divided into two steps, shown in Fig. 5.1.

1. A *depth branch*, which receives only the sensor depth, corrects the errors of the sensor and produces a first estimate called *metric base* and indicated with D_0.
2. A *guidance branch* extracts features from the RGB image and the monocular prior. These features are fused with the features of the depth branch at several scales, and a second decoder predicts a correction that is added to D_0 to obtain the final output $\hat{D}$.

The depth branch never sees the RGB image. As a consequence, the metric base cannot contain texture copied from the image: all the information coming from the guidance is confined to the second step, where it is added as a correction to an estimate that is already metrically reasonable.

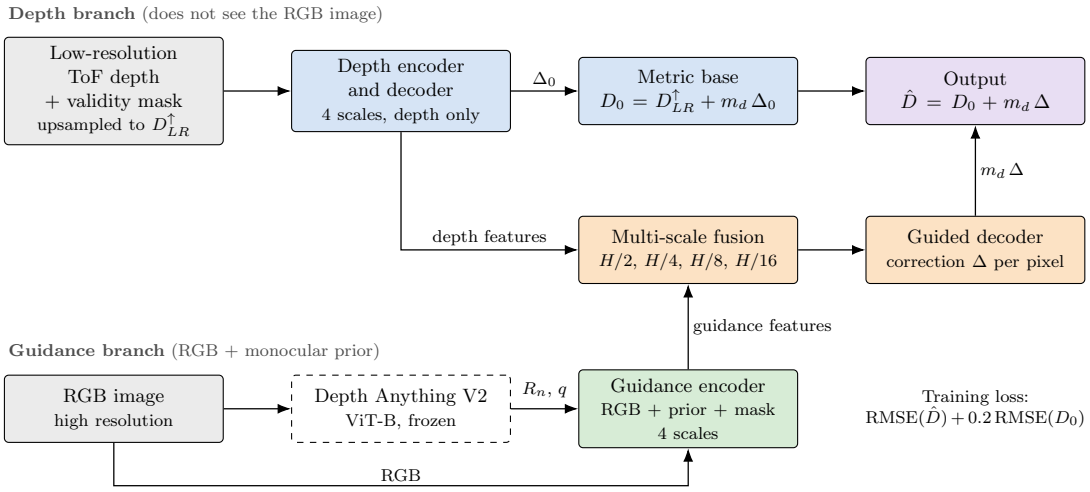


Figure 5.1: Overview of BAF-Net. The depth branch (blue) sees only the sensor depth and produces the metric base D_0. The guidance branch (green) extracts features from the RGB image and from the monocular prior of Depth Anything V2, which is frozen (dashed). The features of the two branches are fused at four scales and a second decoder (orange) predicts the correction Δ that is added to D_0.

5.2 Inputs and Normalization

Let $H \times W$ be the resolution of the RGB image and of the reference depth. The inputs are prepared as follows before being given to the network.

Sensor depth. As described in Section 4.3, the low-resolution depth is resized to $H \times W$ with bicubic interpolation. The result is indicated with $D_{LR}^{\uparrow}$ and is expressed in meters. Its values are limited to the valid range of the dataset, $[0.1, 5.0]$ m. A binary mask M indicates the pixels where the sensor provides a valid measurement.

The depth is then divided by its median value m_d, computed over the valid pixels of the image:

$$\tilde{D} = \frac{D_{LR}^{\uparrow}}{m_d}, \qquad m_d = \text{median}_{p\,:\,M(p)=1}\ D_{LR}^{\uparrow}(p). \tag{5.1}$$

In this way the network always works with values close to one, whether the scene is at one meter or at four meters from the camera. The corrections predicted by the network are expressed in the same relative unit and are multiplied by m_d to return to meters.

Monocular prior. The prior is computed with the ViT-B version of Depth Anything V2. The model is kept frozen and is never trained. Since its output for a given image never changes, the prior is computed once for every image of the dataset and stored on disk.

The model returns a relative inverse depth R, which is related to the true depth only up to an unknown scale and shift (Eq. 2.4). The network should not waste capacity to compensate for this arbitrary scale. For this reason, R is not used directly. Two quantities are derived from it:

- a normalized prior
$$R_n = \frac{R - \text{median}(R)}{p_{95}(R) - p_{05}(R)}, \tag{5.2}$$
where p_{05} and p_{95} are the 5th and 95th percentiles of R in the image. The result is limited to $[-4, 4]$;
- the rank $q \in [0,1]$ of each pixel, which is the fraction of pixels of the image with a smaller value of R. It is computed from 65 quantiles of R, with linear interpolation between them.

Both R_n and q do not change if R is replaced by $\alpha R + \beta$ with $\alpha > 0$. The network is therefore insensitive, by construction, to the unknown scale and shift of the monocular prediction. The two quantities are complementary: R_n keeps the relative size of the depth differences, while q only keeps the ordering of the pixels and spreads them uniformly between 0 and 1.

RGB image. The RGB image is used at full resolution, with values in [0,1].

Padding. The encoders reduce the resolution by a factor of 16. The inputs are therefore extended to a multiple of 16 by repeating the border pixels, and the output is cropped back to $H \times W$.

5.3 Building Blocks

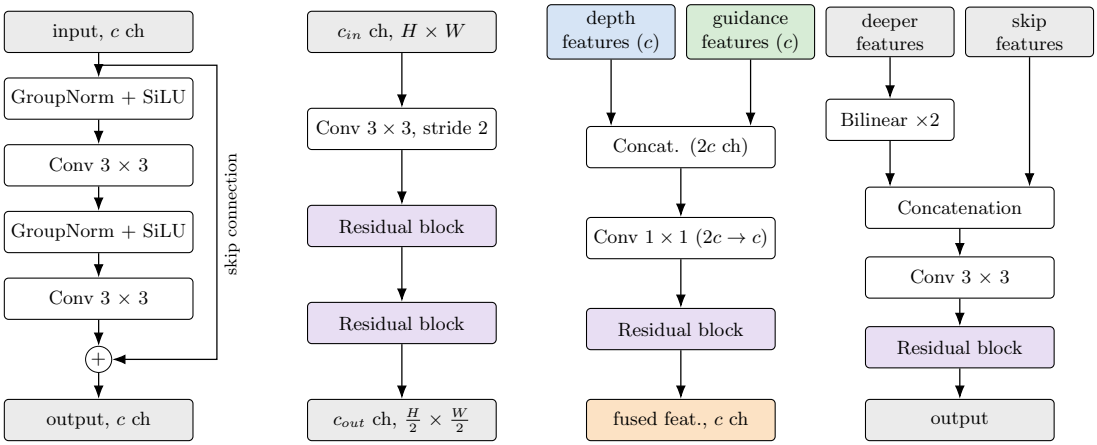
The whole network is built from four simple blocks, shown in Fig. 5.2.

Residual block. The basic unit is a residual block with two 3×3 convolutions. Each convolution is preceded by Group Normalization [60] and by the SiLU activation function [61]. Group Normalization is used instead of Batch Normalization because the network is trained on full images with small batches, where the statistics of the batch are not reliable.

Encoder stage. An encoder stage halves the resolution with a 3×3 convolution with stride 2 and then applies two residual blocks.

Fusion block. A fusion block combines the features of the two branches at the same resolution. The two feature maps are concatenated, a 1×1 convolution brings the number of channels back to the original value, and a residual block processes the result.

Decoder stage. A decoder stage doubles the resolution of the deeper features with bilinear interpolation, concatenates them with the skip features at the same resolution, and applies a 3×3 convolution followed by a residual block. Bilinear interpolation is used instead of transposed convolution because the latter can produce regular checkerboard patterns [62], which in a depth map would appear as invented structure.



(a) Residual block (b) Encoder stage (c) Fusion block (d) Decoder stage

Figure 5.2: Building blocks of BAF-Net. (a) Residual block. (b) Encoder stage, which halves the resolution. (c) Fusion block, which combines depth and guidance features at the same resolution. (d) Decoder stage, which doubles the resolution and uses a skip connection.

5.4 Network Architecture

The complete architecture is shown in Fig. 5.3. It is composed of two encoders with the same structure, a fusion module and two decoders.

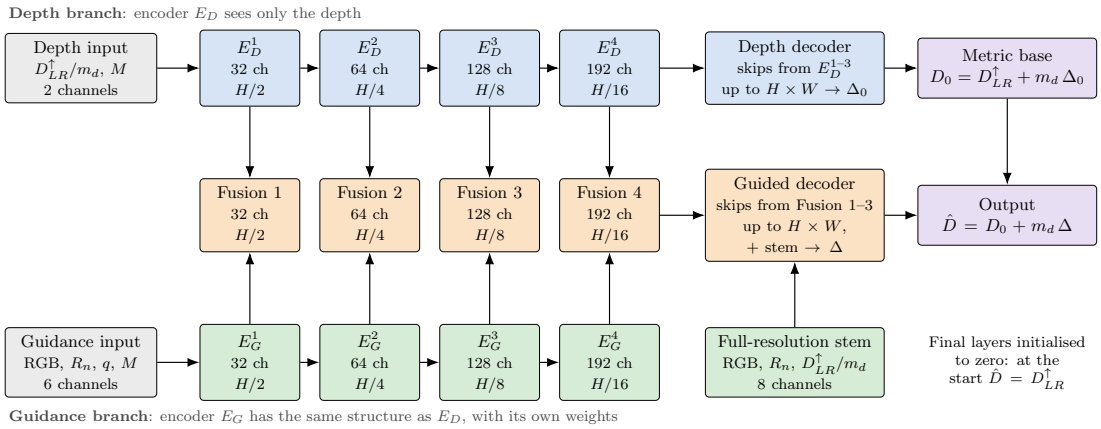


Figure 5.3: Detailed architecture of BAF-Net. The two encoders E_D and E_G have the same structure but separate weights. Their features are fused at each of the four resolutions. The depth decoder uses only the features of E_D, so the metric base D_0 never depends on the RGB image. The guided decoder uses the fused features and, at full resolution, the features of a light stem.

5.4.1 Depth Branch and Metric Base

The depth encoder E_D receives two channels: the normalized depth $\tilde{D}$ and the validity mask M. It is composed of four encoder stages with 32, 64, 128 and 192 channels, which produce features at resolutions $H/2$, $H/4$, $H/8$ and $H/16$.

The depth decoder starts from the deepest features and goes back to $H/2$ with three decoder stages, using the features of E_D as skip connections. The result is brought to full resolution with bilinear interpolation, and a final 3×3 convolution predicts a correction Δ_0 with one channel. The metric base is

$$D_0 = D^{\uparrow}_{LR} + m_d\,\Delta_0. \tag{5.3}$$

The network therefore does not predict the depth from scratch: it predicts how much the sensor measurement must be changed. This is the residual formulation commonly used in super-resolution, where the upsampled input is already close to the solution.

Since this branch only sees the depth, it can only learn what can be understood from the sensor itself: removing noise, correcting systematic errors of the measurements and sharpening the borders that are already visible in the low-resolution depth.

5.4.2 Guidance Branch

The guidance encoder E_G has the same structure as E_D, but its own weights. It receives six channels: the three channels of the RGB image, the normalized prior R_n, the rank q and the validity mask M of the sensor. The mask tells the encoder where the sensor has no measurement, so that the guidance can be used more strongly in those regions.

The RGB image and the monocular prior are given to the same encoder. The two inputs are complementary: the prior indicates where the geometric boundaries are, and the image provides the precise position of the edges at full resolution.

5.4.3 Multi-Scale Fusion and Guided Decoder

At each of the four resolutions, a fusion block combines the features of E_D and E_G. The fusion is performed at all scales because the two types of information are useful at different levels: at low resolution the features describe the general layout of the scene, while at high resolution they describe the exact position of the edges.

The guided decoder has the same structure as the depth decoder, but it works on the fused features. Its output at $H/2$ has 32 channels and is brought to full resolution with bilinear interpolation.

At this point one more source of information is added. The decoder works at $H/2$, so the finest details of the inputs are not directly available to it. A light *stem*, made of a single 3×3 convolution with 8 output channels, is applied at full resolution to the RGB image, the normalized prior R_n and the normalized depth $\tilde{D}$. Its output is concatenated with the features of the decoder, for a total of 40 channels. A final 3×3 convolution predicts the correction Δ, and the output of the network is

$$\hat{D} = D_0 + m_d \, \Delta. \tag{5.4}$$

Both D_0 and $\hat{D}$ are limited to the valid range $[0.1, 5.0]\,\mathrm{m}$.

It is worth noting that the encoder E_D is shared: its features are used both by the depth decoder, to produce D_0, and by the fusion blocks. The RGB image and the prior, however, enter the network only through E_G and the stem, and therefore they can influence the output only through the correction Δ.

5.5 Properties of the Design

The choices described above give the network some properties that hold by construction and do not need to be learned.

The network starts from the input. The two final convolutions, which predict Δ_0 and Δ, are initialized to zero. At the beginning of training, Eq. 5.3 and Eq. 5.4 therefore give $\hat{D} = D_0 = D_{LR}^{\uparrow}$: the network returns exactly the upsampled sensor depth. Training starts from a reasonable solution instead of a random one, and every change with respect to the input must be learned because it reduces the loss.

The metric base does not depend on the RGB image. D_0 is computed only from the sensor depth and its mask. Texture cannot be copied into it, and the contribution of the guidance can be measured directly as the difference $\hat{D} - D_0$.

The output follows the scale of the input. The depth is divided by its median before entering the network, and the corrections are multiplied by the same median. If all the depth values of a scene are multiplied by a constant k, the output is also multiplied by k, apart from the effect of the limits of the valid range. The network does not need to learn separately how to treat near and far scenes.

The unknown scale of the prior has no effect. The prior enters the network only as R_n and q, which do not change under a change of scale and shift of R. The ambiguity of the monocular prediction is therefore removed before the network, and not by the network.

5.6 Loss Function

The network is trained with the following loss:

$$\mathcal{L} = \mathrm{RMSE}\Big(\hat{D}, D_{GT}\Big) + \lambda\, \mathrm{RMSE}(D_0, D_{GT})\,, \qquad \lambda = 0.2. \tag{5.5}$$

The RMSE is computed for each image over the pixels with a valid reference, and then averaged over the images of the batch. This is the same quantity used for the evaluation (Section 4.4), so the network is trained directly on the metric that is reported. Computing the error per image also gives the same weight to every image, regardless of how many valid pixels it contains.

The second term supervises the metric base. Without it, D_0 would be only an intermediate result, and nothing would force it to be a good depth map: the network could leave all the work to the final correction. With this term, the depth branch is required to produce a meaningful estimate by itself. The weight λ is small, so that the final output remains the main objective.

5.7 Implementation and Training Details

Model size. BAF-Net has 7.82 million trainable parameters, divided as shown in Table 5.1. Depth Anything V2 (ViT-B) has 97.5 million parameters, but it is frozen and its output is computed in advance, so it is not part of the trained model.

Table 5.1: Trainable parameters of BAF-Net. The monocular model is frozen and is not included in the total.

Component	Role	Parameters
Depth encoder E_D	depth features	2.42 M
Depth decoder	metric base D_0	0.91 M
Guidance encoder E_G	guidance features	2.42 M
Fusion blocks	fusion at four scales	1.17 M
Guided decoder and stem	output $\hat{D}$	0.91 M
Total (trainable)		7.82 M
Depth Anything V2 (ViT-B)	frozen prior	(97.5 M)

Training. The network is trained on the TOFDSR training set at scale factor $\times 4$, excluding the 500 images reserved for validation (Chapter 4). Training uses full images, without cropping them into patches. The settings are summarized in Table 5.2.

The optimizer is AdamW [63]. The learning rate increases linearly during the first two epochs and then decreases with a cosine schedule down to 5% of its maximum value. The norm of the gradient is limited to 1 to avoid unstable updates, and mixed precision is used to reduce memory and training time.

Table 5.2: Training settings of BAF-Net.

Setting	Value
Optimizer	AdamW, weight decay 10^{-4}
Maximum learning rate	3×10^{-4}
Learning rate schedule	2 warm-up epochs, then cosine decay
Epochs	60
Batch size	8
Input	full images, no patches
Gradient clipping	norm 1.0
Precision	mixed (16/32 bit)
Loss	Eq. 5.5, $\lambda = 0.2$

Data augmentation. Only two simple augmentations are used. The first is a horizontal flip, applied with probability 0.5 to all the inputs and to the reference. The second is a small random change of brightness and contrast, applied only to the RGB image. Vertical flips and rotations are not used, because indoor scenes have a clear vertical direction and rotated scenes would not be realistic. Changes of scale are also avoided, because they would alter the relation between the size of the objects in the image and their distance.

Model selection. After each epoch the model is evaluated on the validation set, and the weights with the lowest validation RMSE are kept. The test set is not used during training or for selecting the model, as described in Section 4.4.1.

Hardware. Training was performed on a single NVIDIA RTX 3090 GPU.

5.8 Summary

BAF-Net separates the reconstruction into two steps. A depth branch, which sees only the sensor, produces a metric base. A guidance branch, which sees the RGB image and a frozen monocular prior, is fused with the depth features at four scales and produces a correction for each pixel. The two final layers start from zero, so that the network initially returns its input, and all quantities are normalized so that the result does not depend on the scale of the scene or on the arbitrary scale of the prior. The next chapter evaluates the model on TOFDSR and measures the contribution of its components.

Chapter 6

Experimental Results

This chapter evaluates BAF-Net on the TOFDSR dataset using real low-resolution depth measurements. The results are compared with those of other GDSR methods, followed by an analysis of the error across the test images. Ablation experiments examine the contributions of RGB guidance and the monocular depth prior. The chapter also compares BAF-Net with direct monocular depth estimation and examines the predicted depth maps, including examples from DORNet.

6.1 Experimental Setup

All experiments follow the real-world protocol described in Chapter 4. TOFDSR is evaluated at scale factor $\times 4$ using low-resolution depth maps acquired by a physical sensor. For each image, RMSE is computed over reference pixels with depths between 0.1 and 5.0 m, after excluding a six-pixel border. The values are then averaged over the dataset and reported in centimeters. The architecture and training procedure are described in Chapter 5.

The validation and test sets have separate roles:

- the *validation set* contains 500 images held out from the training data. It is used for model selection and the ablation experiments;
- the *official test set* contains 560 images and is reserved for evaluating the selected model.

The checkpoint at epoch 46 was selected according to validation RMSE. All test-set results reported here use that checkpoint; neither the model configuration nor its parameters were selected using test-set performance.

The comparison results for other methods are taken from published work. Most are reported in DORNet's evaluation [12], which follows the same real-world protocol. The TPVD result is taken from its original paper [20].

6.2 Comparison with the State of the Art

Table 6.1 reports the published results obtained on the official TOFDSR test set with the real-world ×4 protocol. BAF-Net obtains an RMSE of 4.011 cm, compared with 4.21 cm for DORNet, the best previous result among the listed methods. The difference is approximately 0.20 cm, corresponding to a 4.8% reduction relative to DORNet. The upsampled sensor input has an RMSE of 7.029 cm; BAF-Net reduces this value by approximately 43%.

Table 6.1: Comparison on the official TOFDSR test set using real low-resolution input (×4). RMSE is reported in centimeters (lower is better). The BAF-Net parameter count excludes the frozen monocular model and refers only to its trainable component (Table 5.1).

Method	Venue	Params (M)	RMSE (cm)
DJF [64]	ECCV 2016	0.08	5.84
DJFR [39]	TPAMI 2019	0.08	5.72
CUNet [65]	TPAMI 2020	0.21	6.04
DKN [13]	IJCV 2021	1.16	5.50
FDKN [13]	IJCV 2021	0.69	5.77
FDSR [11]	CVPR 2021	0.60	5.03
DCTNet [14]	CVPR 2022	0.48	5.16
SUFT [66]	ACM MM 2022	22.01	4.37
SFG [43]	AAAI 2024	63.53	4.52
SGNet [15]	AAAI 2024	35.42	4.33
DORNet-T [12]	CVPR 2025	0.46	4.87
TPVD [20]	TPAMI 2025	4.6	4.25
DORNet [12]	CVPR 2025	3.05	4.21
Sensor input (upsampled)	–	–	7.029
BAF-Net (ours)	–	7.82	**4.011**

The relationship between accuracy and trainable parameter count is shown in Figure 6.1. BAF-Net uses 7.82 M trainable parameters, compared with 3.05 M for DORNet. BAF-Net also relies on a frozen monocular model that is excluded from its parameter count. It therefore has a higher computational requirement than the number of trainable parameters alone suggests, and the lower RMSE does not establish an efficiency advantage over DORNet. The trainable part of BAF-Net is

smaller than SUFT, SFG, and SGNet, whose parameter counts range from 22 to 64 million and whose reported RMSE values are above 4.3 cm.

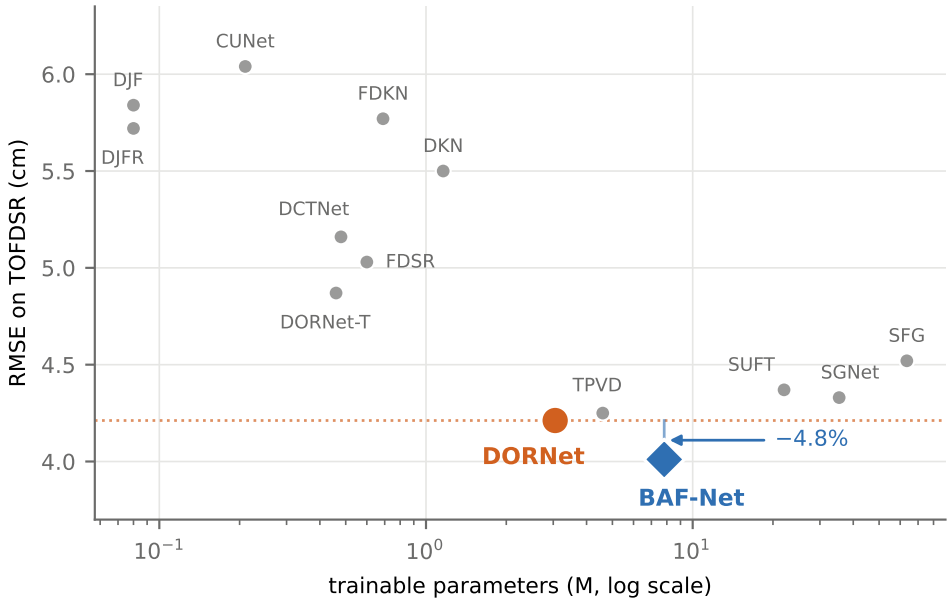


Figure 6.1: Test RMSE on TOFDSR as a function of trainable parameter count, shown on a logarithmic horizontal scale. The dotted line indicates DORNet's RMSE. BAF-Net has the lowest RMSE among the plotted methods.

Lower RMSE values have been reported on TOFDSR under other evaluation settings. DuCos, for example, reports 2.09 cm [17], but its protocol involves synthetic training data and a different preparation of the input depth. As discussed in Section 2.3, these results cannot be compared directly with those of the real-world protocol used here and are not included in Table 6.1.

6.3 Error Distribution

The average test RMSE does not show whether the improvement is consistent between images. Figure 6.2 compares BAF-Net and the sensor input using the RMSE measured on each of the 560 test images.

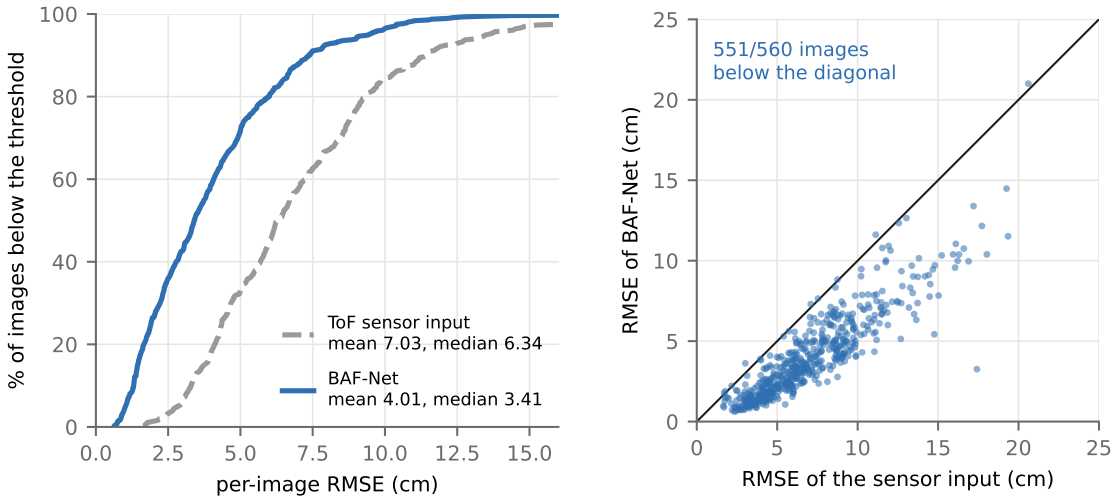


Figure 6.2: RMSE across the 560 test images. Left: cumulative distributions of per-image RMSE for the sensor input and BAF-Net. Right: input and BAF-Net RMSE for each image; points below the diagonal indicate an improvement over the input.

In the cumulative distribution on the left, the BAF-Net curve lies to the left of the input curve over the reported range. The median RMSE decreases from 6.34 cm to 3.41 cm, while the 90th percentile of BAF-Net is 7.38 cm. The right panel gives a direct comparison for each image. BAF-Net has a lower error than the sensor input on 551 of the 560 images (98.4%). Its lower mean RMSE is therefore accompanied by improvements on nearly the entire test set.

6.4 Ablation Study

The ablation experiments evaluate the contributions of the metric reconstruction branch, RGB guidance, and the Depth Anything V2 prior. All results in this section are measured on the 500-image validation set, which was used for model selection. The official test set is not used to compare these configurations.

Table 6.2 includes a model trained with the depth branch alone, without RGB or monocular input. A second configuration retains RGB guidance but disables the Depth Anything V2 prior. The complete BAF-Net receives both RGB and monocular guidance. For the complete network, the table also reports the intermediate metric prediction D_0, before guided refinement.

Table 6.2: Ablation results on 500 validation images, reporting RMSE in centimeters. The RGB-only configuration disables the Depth Anything V2 prior.

Configuration	Validation RMSE (cm)
Sensor input (upsampled)	6.973
Depth branch only (no RGB, no prior)	4.664
RGB guidance (no monocular prior)	4.555
BAF-Net, metric base D_0	4.901
BAF-Net, final output $\hat{D}$	**3.820**

The depth-only model reduces RMSE from 6.973 cm for the input to 4.664 cm. This improvement is achieved without image guidance, indicating that learned reconstruction can correct part of the error present in the sensor measurements. The benefit cannot be attributed entirely to recovering spatial detail lost during downsampling.

Using RGB guidance without the monocular prior lowers RMSE from 4.664 to 4.555 cm, a difference of 0.109 cm (2.3%). With both guidance sources, BAF-Net reaches 3.820 cm, which is 0.735 cm (16.1%) lower than the RGB-guided configuration without the prior. The reduction relative to the depth-only model is 0.844 cm (18.1%). For these configurations, the monocular prior accounts for most of the additional gain from guidance, while RGB alone provides a smaller improvement.

The complete network's intermediate prediction D_0 has an RMSE of 4.901 cm, compared with 4.664 cm for the separately trained depth-only network. These models are optimized differently. In BAF-Net, D_0 is trained with an auxiliary loss weighted by $\lambda = 0.2$ (Eq. 5.5), and the final prediction is optimized through the refinement branch. The refinement reduces the complete model's RMSE from 4.901 to 3.820 cm.

Figure 6.3 shows how the validation RMSE changes during training for the depth-only model, the complete model's intermediate prediction, and its final output. BAF-Net's final output is more accurate than the depth-only baseline from the early epochs, with a difference that persists during training. The lowest validation RMSE occurs at epoch 46, the checkpoint used for the official test evaluation. The RGB-only ablation is included in Table 6.2, but no training curve for it is shown.

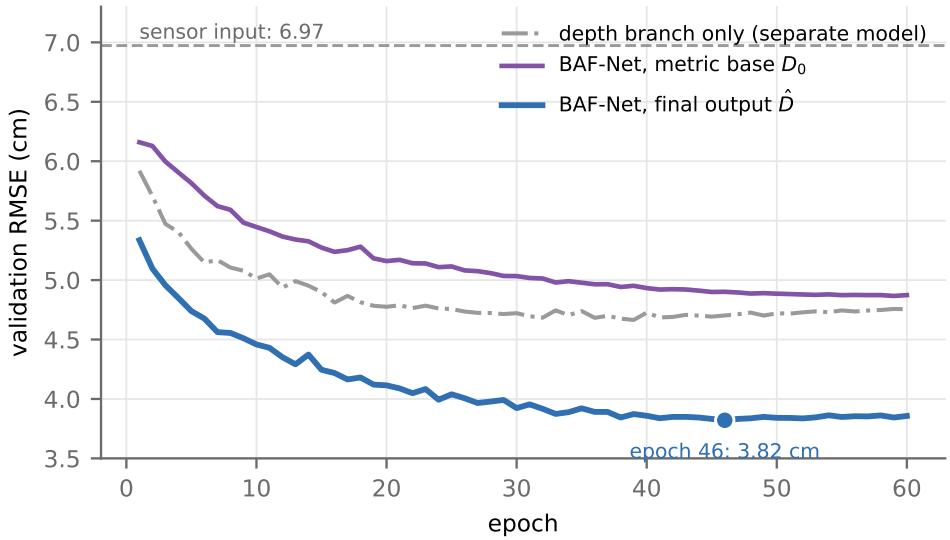


Figure 6.3: Validation RMSE during training for BAF-Net's final prediction $\hat{D}$, its intermediate metric base D_0, and the independently trained depth-only model. The dashed line shows the sensor-input RMSE. The RGB-only ablation is not included in this plot.

6.5 Monocular Prior and BAF-Net

The ablation results show that Depth Anything V2 improves BAF-Net when used as guidance. Its predictions, however, are relative depth estimates and are not necessarily accurate in metric units. To examine this difference, the monocular prediction was evaluated directly as a depth map, without BAF-Net's learned fusion.

Depth Anything V2 predicts relative inverse depth, which must be aligned to metric measurements before evaluation (Section 2.5). For each image, the scale α and shift β in Eq. 2.4 are estimated through a least-squares fit. The fit is then repeated after discarding the 10% of pixels with the largest residuals. Two forms of alignment are considered:

- *aligned to the sensor*: the parameters are fitted to valid sensor measurements, which are available at inference time;
- *aligned to the reference*: the parameters are fitted to the ground-truth depth. This uses information unavailable in a real application and provides an optimistic comparison for global alignment.

This procedure was applied to a random sample of 40 test images. Table 6.3

compares the aligned monocular estimates with the sensor input and BAF-Net on the same sample.

Table 6.3: Comparison of the aligned monocular prediction, sensor input, and BAF-Net on 40 randomly selected test images. DAv2 denotes Depth Anything V2. The last column counts the images on which BAF-Net has a lower RMSE than the listed alternative.

Depth map	Mean (cm)	Median (cm)	BAF-Net better
DAv2, aligned to the sensor	12.87	9.18	40/40
DAv2, aligned to the reference	9.70	8.40	40/40
Sensor input (upsampled)	7.32	6.29	38/40
BAF-Net, metric base D_0	5.25	4.66	–
BAF-Net, final output $\hat{D}$	4.16	3.69	–

The 40-image sample has slightly higher average errors than the full test set. The sensor input has an RMSE of 7.32 cm on the sample, compared with 7.03 cm on all 560 images; the corresponding values for BAF-Net are 4.16 and 4.01 cm. The mean error of the sensor-aligned prior is affected by one image with an RMSE of 120 cm. Excluding this image lowers the mean from 12.87 to 10.12 cm. These values should therefore be interpreted as results for the selected sample, not as estimates of the full test-set performance.

On these images, the monocular prediction has a higher RMSE than the input sensor depth, even when it is aligned using the reference. The predicted object shapes can be well defined while the distances between surfaces or the depth variation within a region remain inaccurate. A global scale and shift cannot correct all these local differences. This explains why BAF-Net uses the monocular output as guidance for refinement while relying on the sensor for metric information.

6.6 Qualitative Results

Figure 6.4 shows the reconstruction stages for a test image at the median improvement over the sensor input. The metric branch reduces RMSE from 6.78 cm to 4.02 cm, largely by correcting the measured depth values. The refinement $\hat{D} - D_0$ is small on many flat surfaces and more visible near object boundaries. After this correction, the RMSE decreases to 3.58 cm.

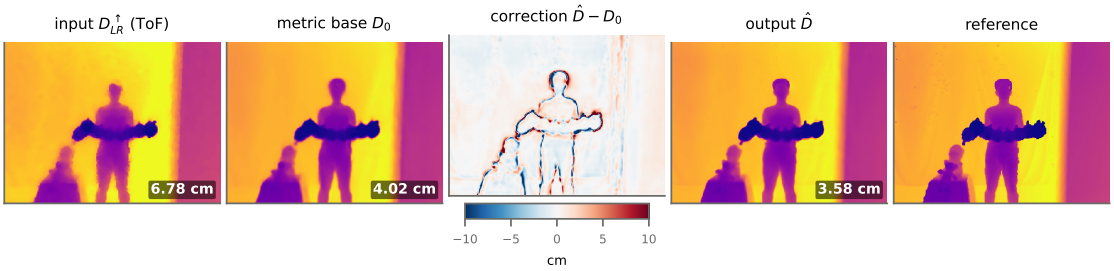


Figure 6.4: Reconstruction stages for the test image at the median improvement. From left to right: upsampled sensor depth, metric base D_0, guidance correction (centimeters), final prediction, and reference. Each panel reports its RMSE.

Figure 6.5 compares the input, sensor-aligned Depth Anything V2 predictions, and BAF-Net results on six test images. Five are selected at the 90th, 70th, 50th, 30th, and 10th percentiles of BAF-Net's improvement over the sensor input. The final row shows the image where its error increases the most. This selection includes examples with different levels of improvement.

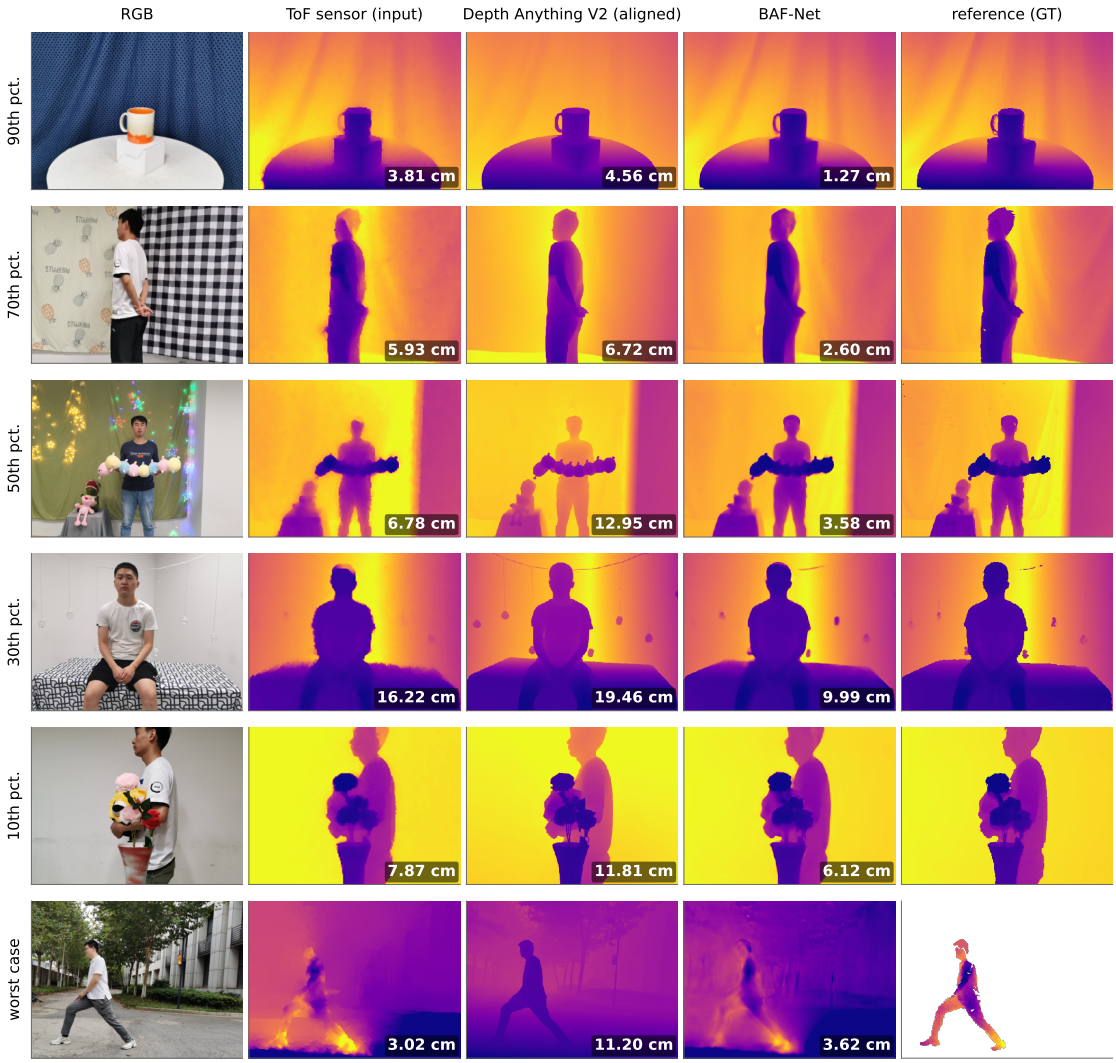


Figure 6.5: Depth reconstructions for five images selected at different percentiles of BAF-Net's improvement, followed by the worst case in the last row (six images in total). From left to right: RGB, sensor input, sensor-aligned Depth Anything V2, BAF-Net, and reference depth. Each row uses a shared color scale. White pixels have no valid reference measurement, and the values in the panels indicate RMSE.

The input depth often has blurred boundaries and visible artifacts around objects. Depth Anything V2 produces more distinct object outlines, but the estimated distances can be inaccurate. In the third row, the outline of the person is well defined even though the aligned prior has almost twice the RMSE of the sensor input. BAF-Net uses this structural information while keeping the reconstructed

depth closer to the reference measurements.

In the last row, the scene is outdoors and the reference is valid mainly over the person. BAF-Net obtains 3.62 cm, compared with 3.02 cm for the input. Because the evaluated region is narrow, errors around the person's contour account for a larger share of the measured RMSE. This is one of the test images where the reconstruction is less accurate than the input.

The signed error maps in Figure 6.6 give a closer view of three examples. The sensor input has positive depth errors across broad areas and larger deviations near object boundaries. The monocular prediction shows errors extending over entire surfaces even after metric alignment. BAF-Net reduces much of the broad error, although deviations remain near depth discontinuities.

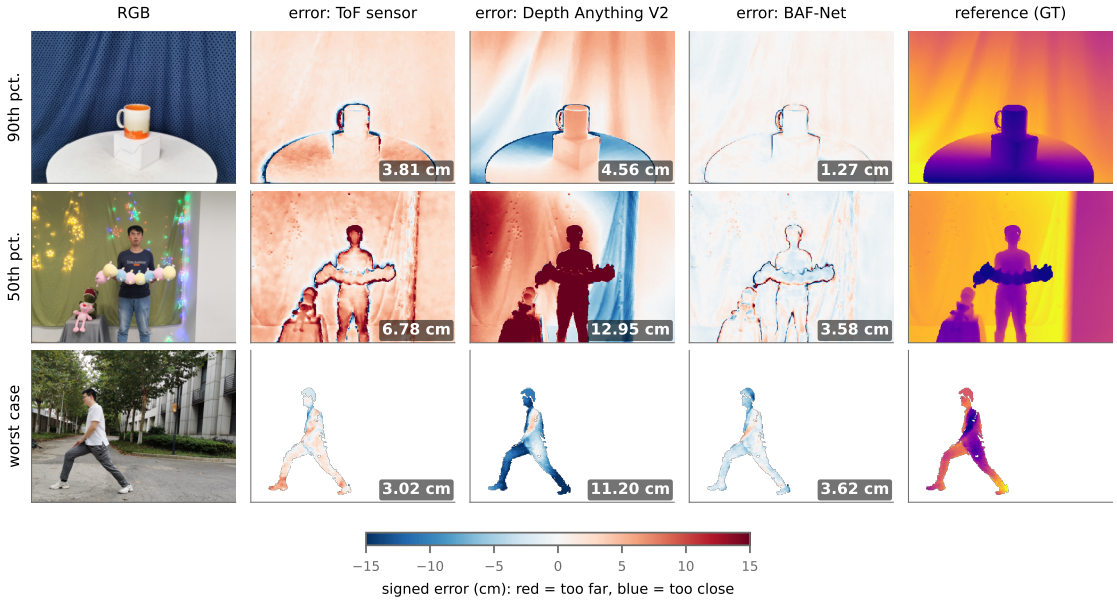


Figure 6.6: Signed depth error (prediction minus reference) for the sensor input, sensor-aligned Depth Anything V2, and BAF-Net. The examples correspond to the 90th percentile, the 50th percentile, and the worst case. Red indicates overestimated distance; blue indicates underestimated distance.

6.6.1 Comparison with DORNet

Figures 6.7 and 6.8 compare BAF-Net with DORNet on five TOFDSR test images. The rows correspond to the 10th, 30th, 50th, 70th, and 90th percentiles of BAF-Net's improvement over the sensor input. The first figure shows the reconstructed depth maps alongside the ToF measurements and the reference. The second shows

the signed errors for the same images. The RMSE reported in each panel allows the two predictions to be compared on individual examples.

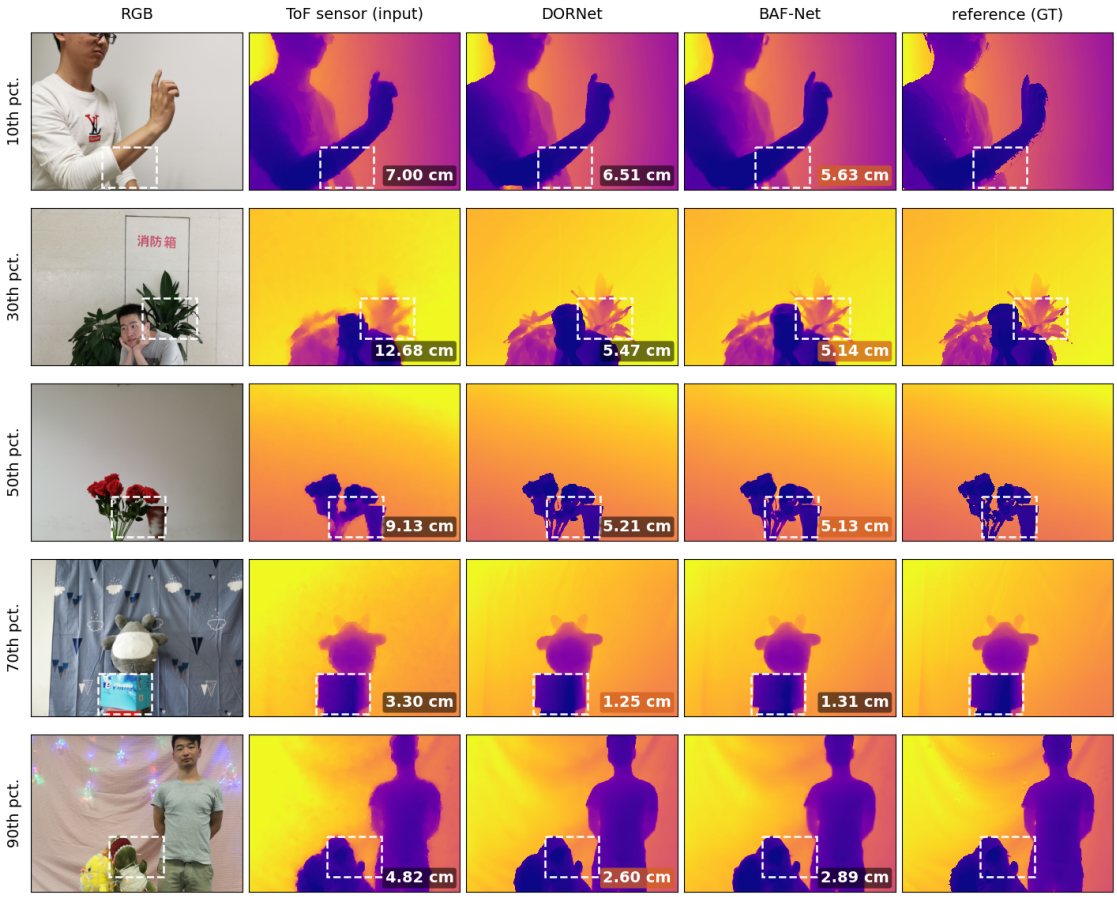


Figure 6.7: Depth comparison on five TOFDSR test images, selected at the 10th, 30th, 50th, 70th, and 90th percentiles of BAF-Net's improvement over the input. Columns: RGB, ToF input, DORNet, BAF-Net, and reference. Dashed boxes mark selected regions; values are per-image RMSE (cm).

For the first three images, BAF-Net has a lower RMSE than DORNet: 5.63 versus 6.51 cm for the person with a raised arm, 5.14 versus 5.47 cm for the scene with plants, and 5.13 versus 5.21 cm for the flowers. DORNet performs slightly better on the remaining two images, with 1.25 versus 1.31 cm for the toy and 2.60 versus 2.89 cm for the standing person. The relative accuracy therefore changes between scenes, even though BAF-Net has the lower RMSE over the full test set (Table 6.1).

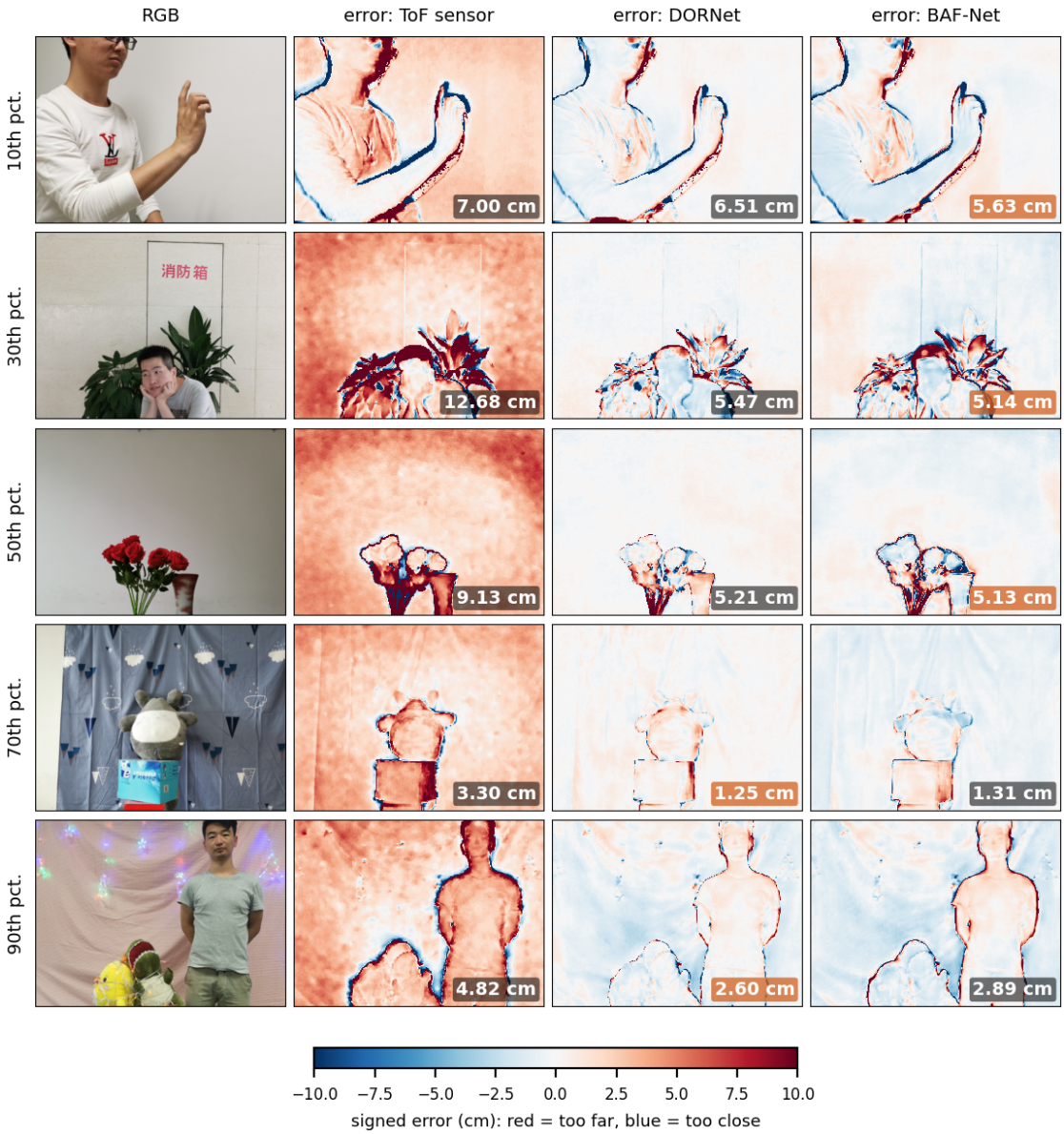


Figure 6.8: Signed depth errors for the five images in Figure 6.7. Columns: RGB and errors of the ToF input, DORNet, and BAF-Net. Red indicates overestimated depth and blue underestimated depth (−10 to 10 cm). Values are per-image RMSE.

The error maps show that both networks remove much of the broad error visible in the input depth. Their remaining errors are often concentrated around object contours, particularly the arm, leaves, and flowers. Some local differences are visible between their predictions, but neither method is consistently more accurate in every highlighted region. These examples illustrate the types of errors left by

the two models and complement the aggregate RMSE comparison.

6.7 Summary

On the official TOFDSR test set, BAF-Net obtains an RMSE of 4.011 cm, compared with 4.21 cm reported for DORNet using the same real-world protocol. BAF-Net reduces the input error on 551 of the 560 test images. The validation ablations show that the depth-only model reaches 4.664 cm and that RGB guidance alone lowers this to 4.555 cm. Adding the Depth Anything V2 prior brings the RMSE to 3.820 cm. The direct evaluation of the aligned monocular predictions shows that they are less accurate in metric terms than the sensor input on the selected 40 images. The qualitative comparison with DORNet also shows that the two methods retain errors around depth boundaries and that the better method can vary from one image to another.

Bibliography

[1] Daniel Scharstein and Richard Szeliski. «A Taxonomy and Evaluation of Dense Two-Frame Stereo Correspondence Algorithms». In: *International Journal of Computer Vision* 47 (2002), pp. 7–42. DOI: 10.1023/A:1014573219977 (cit. on p. 2).

[2] Kourosh Khoshelham and Sander Oude Elberink. «Accuracy and Resolution of Kinect Depth Data for Indoor Mapping Applications». In: *Sensors* 12.2 (2012), pp. 1437–1454. DOI: 10.3390/s120201437 (cit. on p. 2).

[3] Sergi Foix, Guillem Alenyà, and Carme Torras. «Lock-in Time-of-Flight (ToF) Cameras: A Survey». In: *IEEE Sensors Journal* 11.9 (2011), pp. 1917–1926. DOI: 10.1109/JSEN.2010.2101060 (cit. on p. 3).

[4] Jaesung Choe, Sunghoon Im, Francois Rameau, Minjun Kang, and In So Kweon. «VolumeFusion: Deep Depth Fusion for 3D Scene Reconstruction». In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2021, pp. 16086–16095 (cit. on p. 4).

[5] Oleg Voynov, Alexey Artemov, Vage Egiazarian, Alexander Notchenko, Gleb Bobrovskikh, Evgeny Burnaev, and Denis Zorin. «Perceptual Deep Depth Super-Resolution». In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2019, pp. 5653–5663 (cit. on p. 4).

[6] Chao Liu, Jinwei Gu, Kihwan Kim, Srinivasa G. Narasimhan, and Jan Kautz. «Neural RGB→D Sensing: Depth and Uncertainty From a Video Camera». In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019, pp. 10986–10995 (cit. on p. 4).

[7] Xiaokang Chen, Kwan-Yee Lin, Jing Wang, Wei Wu, Chen Qian, Hongsheng Li, and Gang Zeng. «Bi-Directional Cross-Modality Feature Propagation with Separation-and-Aggregation Gate for RGB-D Semantic Segmentation». In: *European Conference on Computer Vision*. 2020, pp. 561–577 (cit. on p. 4).

[8] Hao Zhou, Lu Qi, Hai Huang, Xu Yang, Zhen Wan, and Xiang Wen. «CANet: Co-Attention Network for RGB-D Semantic Segmentation». In: *Pattern Recognition* 124 (2022), p. 108468 (cit. on p. 4).

[9] Runmin Cong, Jianjun Lei, Huazhu Fu, Junwei Hou, Qingming Huang, and Sam Kwong. «Going from RGB to RGBD Saliency: A Depth-Guided Transformation Model». In: *IEEE Transactions on Cybernetics* 50.8 (2020), pp. 3627–3639 (cit. on p. 4).

[10] Tak-Wai Hui, Chen Change Loy, and Xiaoou Tang. «Depth Map Super-Resolution by Deep Multi-Scale Guidance». In: *Proceedings of the European Conference on Computer Vision (ECCV)*. 2016, pp. 353–369 (cit. on pp. 5, 14).

[11] Lingzhi He, Hongguang Zhu, Feng Li, Huihui Bai, Runmin Cong, Chunjie Zhang, Chunyu Lin, Meiqin Liu, and Yao Zhao. «Towards Fast and Accurate Real-World Depth Super-Resolution: Benchmark Dataset and Baseline». In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2021, pp. 9229–9238 (cit. on pp. 5, 14–16, 19, 29, 48).

[12] Zhengxue Wang, Zhiqiang Yan, Jinshan Pan, Guangwei Gao, Kai Zhang, and Jian Yang. «DORNet: A Degradation Oriented and Regularized Network for Blind Depth Super-Resolution». In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2025, pp. 15813–15822 (cit. on pp. 5, 6, 15, 16, 19, 29, 35, 36, 48).

[13] Beomjun Kim, Jean Ponce, and Bumsub Ham. «Deformable Kernel Networks for Joint Image Filtering». In: *International Journal of Computer Vision* 129.2 (2021), pp. 579–600 (cit. on pp. 5, 12, 14, 48).

[14] Zixiang Zhao, Jiangshe Zhang, Shuang Xu, Zudi Lin, and Hanspeter Pfister. «Discrete Cosine Transform Network for Guided Depth Map Super-Resolution». In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022, pp. 5697–5707 (cit. on pp. 5, 14, 48).

[15] Zhengxue Wang, Zhiqiang Yan, and Jian Yang. «SGNet: Structure Guided Network via Gradient-Frequency Awareness for Depth Map Super-resolution». In: *Proceedings of the AAAI Conference on Artificial Intelligence* 38.6 (2024), pp. 5823–5831. DOI: 10.1609/aaai.v38i6.28395 (cit. on pp. 5, 11, 14, 48).

[16] Nando Metzger, Rodrigo Caye Daudt, and Konrad Schindler. «Guided Depth Super-Resolution by Deep Anisotropic Diffusion». In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2023, pp. 18237–18246 (cit. on pp. 5, 14).

[17] Zhiqiang Yan, Zhengxue Wang, Haoye Dong, Jun Li, Jian Yang, and Gim Hee Lee. «DuCos: Duality Constrained Depth Super-Resolution via Foundation Model». In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2025, pp. 8361–8371 (cit. on pp. 6, 15, 18, 49).

[18] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. «Depth Anything V2». In: *Advances in Neural Information Processing Systems*. Vol. 37. 2024. DOI: 10.52202/079017-0688 (cit. on pp. 6, 17, 37).

[19] Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. «Colored Point Cloud Registration Revisited». In: *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 2017, pp. 143–152. DOI: 10.1109/ICCV.2017.25 (cit. on pp. 6, 7, 20).

[20] Zhiqiang Yan, Kun Wang, Xiang Li, Guangwei Gao, Jun Li, and Jian Yang. «Tri-Perspective View Decomposition for Geometry Aware Depth Completion and Super-Resolution». In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 47.12 (2025), pp. 11330–11347. DOI: 10.1109/TPAMI.2025.3596391 (cit. on pp. 6, 7, 34, 35, 48).

[21] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. «Deep Learning». In: *Nature* 521.7553 (2015), pp. 436–444 (cit. on p. 8).

[22] Diederik P. Kingma and Jimmy Ba. «Adam: A Method for Stochastic Optimization». In: *International Conference on Learning Representations (ICLR)*. 2015 (cit. on p. 9).

[23] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. «U-Net: Convolutional Networks for Biomedical Image Segmentation». In: *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2015 (cit. on p. 10).

[24] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. «Deep Residual Learning for Image Recognition». In: *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016 (cit. on p. 10).

[25] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. «Attention Is All You Need». In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2017 (cit. on p. 10).

[26] Alexey Dosovitskiy et al. «An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale». In: *International Conference on Learning Representations (ICLR)*. 2021 (cit. on p. 10).

[27] Yochai Blau and Tomer Michaeli. «The Perception-Distortion Tradeoff». In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2018 (cit. on p. 11).

[28] Zhiwei Zhong, Xianming Liu, Junjun Jiang, Debin Zhao, and Xiangyang Ji. «Guided Depth Map Super-Resolution: A Survey». In: *ACM Computing Surveys* 55.14s (2023), pp. 1–36. DOI: 10.1145/3584860 (cit. on pp. 11–14).

[29] Johannes Kopf, Michael F. Cohen, Dani Lischinski, and Matt Uyttendaele. «Joint Bilateral Upsampling». In: *ACM Transactions on Graphics* 26.3 (2007) (cit. on p. 12).

[30] Kaiming He, Jian Sun, and Xiaoou Tang. «Guided Image Filtering». In: *European Conference on Computer Vision*. 2010, pp. 1–14 (cit. on p. 12).

[31] Bumsub Ham, Minsu Cho, and Jean Ponce. «Robust Guided Image Filtering Using Nonconvex Potentials». In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 40.1 (2018), pp. 192–207 (cit. on p. 12).

[32] James Diebel and Sebastian Thrun. «An Application of Markov Random Fields to Range Sensing». In: *Advances in Neural Information Processing Systems*. 2005, pp. 291–298 (cit. on p. 13).

[33] David Ferstl, Christian Reinbacher, Rene Ranftl, Matthias Rüther, and Horst Bischof. «Image Guided Depth Upsampling Using Anisotropic Total Generalized Variation». In: *Proceedings of the IEEE International Conference on Computer Vision*. 2013, pp. 993–1000 (cit. on p. 13).

[34] Si Lu, Xiaofeng Ren, and Feng Liu. «Depth Enhancement via Low-Rank Matrix Completion». In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2014, pp. 3390–3397. DOI: 10.1109/CVPR.2014.433 (cit. on pp. 13, 18).

[35] Xianming Liu, Dong Zhai, Rong Chen, Xiangyang Ji, Debin Zhao, and Wen Gao. «Depth Super-Resolution via Joint Color-Guided Internal and External Regularizations». In: *IEEE Transactions on Image Processing* 28.4 (2019), pp. 1636–1645 (cit. on p. 13).

[36] Ivana Tosic and Stefan Drewes. «Learning Joint Intensity-Depth Sparse Representations». In: *IEEE Transactions on Image Processing* 23.5 (2014), pp. 2122–2132 (cit. on p. 13).

[37] Peipei Song, Xin Deng, João F. C. Mota, Nikos Deligiannis, Pier Luigi Dragotti, and Miguel R. D. Rodrigues. «Multimodal Image Super-Resolution via Joint Sparse Representations Induced by Coupled Dictionaries». In: *IEEE Transactions on Computational Imaging* 6 (2019), pp. 57–72 (cit. on p. 13).

[38] Fei Gao, Xin Deng, Meng Xu, Jing Xu, and Pier Luigi Dragotti. «Multi-Modal Convolutional Dictionary Learning». In: *IEEE Transactions on Image Processing* 31 (2022), pp. 1325–1339 (cit. on p. 13).

[39] Yijun Li, Jia-Bin Huang, Narendra Ahuja, and Ming-Hsuan Yang. «Joint Image Filtering with Deep Convolutional Networks». In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41.8 (2019), pp. 1909–1923 (cit. on pp. 14, 48).

[40] Riccardo de Lutio, Alexander Becker, Stefano D'Aronco, Stefania Russo, Jan Dirk Wegner, and Konrad Schindler. «Learning Graph Regularisation for Guided Super-Resolution». In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022, pp. 1979–1988 (cit. on p. 14).

[41] Ilan Ariav and Israel Cohen. «Depth Map Super-Resolution via Cascaded Transformers Guidance». In: *Frontiers in Signal Processing* 2 (2022) (cit. on p. 14).

[42] Tayyab Nasir, Daochang Liu, and Ajmal Mian. *WAVE: Reversing the Guidance Hierarchy for Coarse-to-Fine Guided Depth Super-Resolution*. 2026. arXiv: 2608.25302 [cs.CV] (cit. on pp. 15, 18).

[43] Jiayi Yuan, Haobo Jiang, Xiang Li, Jianjun Qian, Jun Li, and Jian Yang. «Structure Flow-Guided Network for Real Depth Super-resolution». In: *Proceedings of the AAAI Conference on Artificial Intelligence* 37.3 (2023), pp. 3340–3348. DOI: 10.1609/aaai.v37i3.25441 (cit. on pp. 16, 48).

[44] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. «Towards Robust Monocular Depth Estimation: Mixing Datasets for Zero-Shot Cross-Dataset Transfer». In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44.3 (2022), pp. 1623–1637 (cit. on p. 17).

[45] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. «Vision Transformers for Dense Prediction». In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2021 (cit. on p. 17).

[46] Lihe Yang, Bingyi Kang, Zilong Huang, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. «Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data». In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024 (cit. on p. 17).

[47] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. «DINOv2: Learning Robust Visual Features without Supervision». In: *Transactions on Machine Learning Research* (2024) (cit. on p. 17).

[48] Zehan Wang, Siyu Chen, Lihe Yang, Jialei Wang, Ziang Zhang, Hengshuang Zhao, and Zhou Zhao. *Depth Anything with Any Prior*. 2025. arXiv: 2505.10565 [cs.CV] (cit. on p. 18).

[49] Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus. «Indoor Segmentation and Support Inference from RGBD Images». In: *Proceedings of the European Conference on Computer Vision (ECCV)*. 2012, pp. 746–760 (cit. on p. 18).

[50] Daniel Scharstein and Richard Szeliski. «High-Accuracy Stereo Depth Maps Using Structured Light». In: *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*. Vol. 1. 2003, pp. 195–202. DOI: 10.1109/CVPR.2003.1211354 (cit. on p. 18).

[51] Zhiqiang Yan, Yuankai Lin, Kun Wang, Yupeng Zheng, Yufei Wang, Zhenyu Zhang, Jun Li, and Jian Yang. «Tri-Perspective View Decomposition for Geometry-Aware Depth Completion». In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 4874–4884 (cit. on p. 19).

[52] Jaehoon Cho, Dongbo Min, Youngjung Kim, and Kwanghoon Sohn. «DIML/CVL RGB-D Dataset: 2M RGB-D Images of Natural Indoor and Outdoor Scenes». In: *arXiv preprint arXiv:2110.11590* (2021) (cit. on p. 19).

[53] Mike Roberts, Jason Ramapuram, Anurag Ranjan, Atulit Kumar, Miguel Angel Bautista, Nathan Paczan, Russ Webb, and Joshua M. Susskind. «Hypersim: A Photorealistic Synthetic Dataset for Holistic Indoor Scene Understanding». In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2021, pp. 10912–10922. DOI: 10.1109/ICCV48922.2021.01073 (cit. on p. 19).

[54] Brian Curless and Marc Levoy. «A Volumetric Method for Building Complex Models from Range Images». In: *Proceedings of SIGGRAPH*. 1996 (cit. on p. 23).

[55] Richard A. Newcombe et al. «KinectFusion: Real-Time Dense Surface Mapping and Tracking». In: *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2011 (cit. on p. 23).

[56] Radu Bogdan Rusu, Nico Blodow, and Michael Beetz. «Fast Point Feature Histograms (FPFH) for 3D Registration». In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2009 (cit. on p. 23).

[57] Qian-Yi Zhou, Jaesik Park, and Vladlen Koltun. «Fast Global Registration». In: *European Conference on Computer Vision (ECCV)*. 2016 (cit. on p. 23).

[58] Paul J. Besl and Neil D. McKay. «A Method for Registration of 3-D Shapes». In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 14.2 (1992), pp. 239–256 (cit. on p. 23).

[59] Yang Chen and Gérard Medioni. «Object Modelling by Registration of Multiple Range Images». In: *Image and Vision Computing* 10.3 (1992), pp. 145–155 (cit. on p. 23).

[60] Yuxin Wu and Kaiming He. «Group Normalization». In: *European Conference on Computer Vision (ECCV)*. 2018 (cit. on p. 40).

[61] Stefan Elfwing, Eiji Uchibe, and Kenji Doya. «Sigmoid-Weighted Linear Units for Neural Network Function Approximation in Reinforcement Learning». In: *Neural Networks* 107 (2018), pp. 3–11 (cit. on p. 40).

[62] Augustus Odena, Vincent Dumoulin, and Chris Olah. «Deconvolution and Checkerboard Artifacts». In: *Distill* (2016) (cit. on p. 40).

[63] Ilya Loshchilov and Frank Hutter. «Decoupled Weight Decay Regularization». In: *International Conference on Learning Representations (ICLR)*. 2019 (cit. on p. 45).

[64] Yijun Li, Jia-Bin Huang, Narendra Ahuja, and Ming-Hsuan Yang. «Deep Joint Image Filtering». In: *European Conference on Computer Vision (ECCV)*. 2016 (cit. on p. 48).

[65] Xin Deng and Pier Luigi Dragotti. «Deep Convolutional Neural Network for Multi-Modal Image Restoration and Fusion». In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43.10 (2021), pp. 3333–3348 (cit. on p. 48).

[66] Wuxuan Shi, Mang Ye, and Bo Du. «Symmetric Uncertainty-Aware Feature Transmission for Depth Super-Resolution». In: *ACM International Conference on Multimedia (ACM MM)*. 2022 (cit. on p. 48).